FINITENESS AND SHRINKING SPECTRAL WINDOWS FOR MAASS EULER PRODUCTS

ANDREW R. BOOKER, MIN LEE, AND NINA ZUBRILINA

Written by GPT-6 Astra

Abstract. This companion to *An asymptotic count of Euler products at fixed conductor* develops the weight-zero spectral-window argument. For every fixed admissible level, the number of solutions at a fixed spectral parameter ν is $O_N(1+|\nu|)$, including the complementary range. We prove this from the strict spanning certificate, the original functional equation, and a uniform Bessel comparison. For each fixed parity, the number of solutions with parameter in $[t, t+\varepsilon(t)]$ is asymptotic to $\nu(N)t\varepsilon(t)/12$, provided $\varepsilon(t)\log\log\log t \to \infty$. The lower bound uses a quantitative rank-one adaptation of the Lindenstrauss–Venkatesh construction. The upper bound separates the exact genuine spectrum from the excess in a common weighted space and controls the latter by the regular residual filter.

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The squarefree-conductor assertion in the accompanying paper is a special case of the admissible-level results proved here.

1. Statement and counting convention

Fix an admissible level N, meaning $16 \nmid N$ and no odd prime square divides N. Let $\mathscr{E}_N^\sigma(\nu)$ denote the set of normalized complex coefficient sequences $a = (a_n)_{n\ge1}$, $a_1 = 1$, with $\nu \in \mathbb{R} \cup i(-1/2, 1/2)$ and $\sigma \in \{0, 1\}$, satisfying

$$L_a(s) = \sum_{n\ge1} a_n n^{-s} = \prod_p (1 - a_p p^{-s} + \xi_N(p) p^{-2s})^{-1} \qquad (\Re s \gg 1),$$

where $\xi_N(p) = \mathbf{1}_{p\nmid N}$, and $a_p^2 = p^{-1}$ for $p \parallel N$, while $a_p = 0$ for $p^2 \mid N$. The completed function

$$\Lambda_a(s) = N^{s/2}\Gamma_{\mathbb{R}}(s + \sigma + i\nu)\Gamma_{\mathbb{R}}(s + \sigma - i\nu)L_a(s), \qquad \Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2),$$

is required to be entire of finite order and to satisfy $\Lambda_a(s) = \epsilon_a \Lambda_a(1 - s)$ for some $\epsilon_a \in \{\pm1\}$. These are precisely the untwisted Euler-product hypotheses; no twisted functional equation or modularity is assumed. Using $N^{(s-1/2)/2}$ instead changes the completion only by a nonzero constant. The real spectral parameter is counted with $r \ge 0$, so that the Laplace eigenvalue is $1/4 + r^2$. Parity σ corresponds to reflection eigenvalue $(-1)^\sigma$. Let $\mathscr{E}_N^\sigma([t, t+\varepsilon])$ be the set of pairs (r, a) with $r \in [t, t+\varepsilon]$ and $a \in \mathscr{E}_N^\sigma(r)$, summed over Fricke and ramified signs. The arithmetic factor is multiplicative and, at admissible levels, is

$$\nu(N) = \varphi(N) \quad (N \text{ squarefree}), \qquad \nu(4m) = \varphi(m), \qquad \nu(8m) = 3\varphi(m) \quad (m \text{ odd squarefree}).$$

Theorem 1.1. *For every admissible N there is a constant C_N such that, for both parities and every $\nu \in \mathbb{R} \cup i(-1/2, 1/2)$,*

$$\#\mathscr{E}_N^\sigma(\nu) \le C_N(1 + |\nu|).$$

In particular each of these sets is finite. More strongly, for a fixed choice of Ramanujan-twist root phases on the divisors of a suitable integer Q_N, the corresponding linear solution space has dimension $O_N(1 + |\nu|)$ and is determined by its first $O_N(1 + |\nu|)$ coefficients.

The proof of Theorem 1.1 is given in Section 2. The next theorem refines its real-parameter bound and counts solutions while the parameter varies.

Theorem 1.2. *Fix an admissible level N and a parity $\sigma \in \{0, 1\}$. Suppose $0 < \varepsilon(t) \le 1$ satisfies $\varepsilon(t) \log\log\log t \to \infty$. Then*

$$\#\mathscr{E}_N^\sigma([t, t + \varepsilon(t)]) = \frac{\nu(N)}{12} t\varepsilon(t) + o_N(t\varepsilon(t)).$$

Summing the two parities gives leading coefficient $\nu(N)/6$. In particular the number of solutions at an individual real parameter r is $o_N(r)$ as $r \to \infty$.

This note uses the strict reconstruction certificate, the original Ramanujan functional equations, and the quantitative regular residual filter from the companion holomorphic manuscript. Its new tasks are to count varying Laplace parameters in a common space and to control the zero Fourier channel and the cusps. Distinct parameters need not give orthogonal vectors in the reconstruction norm.

For $N > 1$ the full automorphic newspace at an admissible level is cuspidal. Indeed an Eisenstein representation with trivial central character is induced from χ and χ^{-1}, and its finite conductor is the square of the conductor of χ. Admissibility forces this conductor to be one: the only other possible divisor would be two, and there is no primitive character of conductor two. Thus every Eisenstein constituent is old from level one. The constant representation is old as well. This observation identifies the full new/sign target projection used below with the cuspidal target in the counting lemma. At $N = 1$, periodicity and the Fricke equation give $F|T = F$ and $F|S = wF$. For nonzero F, the projective relations $(ST)^3 = S^2 = I$ force $w = 1$, and hence give modularity under the generators T, S of the full modular group. Vanishing of the constant term makes the solution cuspidal, and the

ordinary genuine spectral count applies directly. The operator argument below may therefore be restricted to $N > 1$.

The weighted wave operator acts by $h(r)$ on an actual solution even though its unweighted realization on the translation cylinder need not be square integrable. This follows from finite propagation: on each compact set one may replace the smooth Laplace eigenfunction by compactly supported data agreeing on its propagation neighborhood. Local uniqueness for the wave equation then gives the eigenfunction identity. The bounded weighted wave operator extends this local identity to the reconstructed weighted space.

2. Finiteness at a fixed spectral parameter

This section proves the fixed-parameter bound independently of the spectral-window asymptotic. It also supplies the finite-dimensional solution spaces used later. The argument is the analytic counterpart of the proof formalized in `MaassEulerFinitenessV1.lean`; its intermediate statements are recorded below so that no finiteness or modularity conclusion is being used as an input.

2.1. The analytic solution spaces and the Ramanujan equations.

Write

$$\mathcal{P} = \mathbb{R} \cup i(-\tfrac{1}{2}, \tfrac{1}{2}), \qquad \lambda_\nu = \tfrac{1}{4} + \nu^2 \quad (\nu \in \mathcal{P}).$$

Thus λ_ν is real and positive, including in the complementary range. For a sequence $a = (a_n)_{n\ge1}$ of polynomial growth put $L_a(s) = \sum_{n\ge1} a_n n^{-s}$. Polynomial growth is an individual condition; a common growth exponent is not assumed for the solution space. The coefficient a_1 is unrestricted in this linear space; normalization is imposed only on the Euler products being counted. Convergence of the original Dirichlet series at any point already implies such a bound, since its terms tend to zero. Let

$$c_q(n) = \sum_{\substack{u \bmod q \\ (u,q)=1}} e^{2\pi i n u/q}, \quad \ell_N(q) = \operatorname{lcm}(N, q^2), \quad L_a(s, c_q) = \sum_{n\ge1} a_n c_q(n) n^{-s}.$$

For fixed ν, σ and a choice of phases ϵ_q of absolute value one, let $\mathcal{F}_{\nu,\sigma,\epsilon}(N; Q)$ be the complex vector space of such sequences for which, for every $q \mid Q$,

$$\Lambda_a(s, c_q) = \ell_N(q)^{s/2} \Gamma_{\mathbb{R}}(s + \sigma + i\nu) \Gamma_{\mathbb{R}}(s + \sigma - i\nu) L_a(s, c_q) \tag{2.1}$$

extends to an entire function of finite order and satisfies $\Lambda_a(s, c_q) = \epsilon_q \Lambda_a(1 - s, c_q)$. Here $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$. Multiplying a completion by the constant $\ell_N(q)^{-1/4}$ makes no difference to its functional equation. Finite order and polynomial growth are preserved by finite linear combinations, so this is indeed a vector space.

For the Euler products in the statement, all these twisted equations follow from the untwisted equation. We give the calculation, since these equations are consequences rather than additional hypotheses. Put $\xi_N(p) = 1$ if $p \nmid N$ and 0 otherwise, and

$$D_{p,\alpha}(s) = (1 - a_p p^{-s} + \xi_N(p) p^{-2s}) \sum_{j\ge0} a_{p^j} c_{p^\alpha}(p^j) p^{-js}.$$

Multiplicativity gives $L_a(s, c_q) = L_a(s) \prod_{p^\alpha \| q} D_{p,\alpha}(s)$. The prime-power recurrence gives, with $a_{p^j} = 0$ for $j < 0$,

$$\begin{aligned}
p \nmid N: \quad & D_{p,\alpha}(s) = p^{\alpha(1-s)} \big(a_{p^\alpha} + p^{-1} a_{p^{\alpha-2}} - a_{p^{\alpha-1}} (p^{-1+s} + p^{-s})\big), \\
& D_{p,\alpha}(s) = p^{\alpha(1-2s)} D_{p,\alpha}(1 - s), \\
p \mid N: \quad & D_{p,\alpha}(s) = p^{(\alpha-1)(1-s)} a_p^{\alpha-1} (-1 + a_p p^{1-s}).
\end{aligned}$$

If $p \parallel N$, the last expression transforms by the factor $-a_p\sqrt{p}\,p^{(2\alpha-1)(1/2-s)}$. If $p^2 \mid N$, then $a_p = 0$: the factor is -1 when $\alpha = 1$ and is identically zero when $\alpha \geq 2$. Combining these identities with the conductor in (2.1) proves the twisted equation with

$$\epsilon_q = \epsilon_a \prod_{\substack{p\mid(q,N)\\ p\parallel N}} (-a_p\sqrt{p}) \in \{\pm 1\}. \tag{2.2}$$

The extra factors are finite Dirichlet polynomials, so entire continuation and finite order are retained. An identically zero twist satisfies its equation with the same prescribed sign.

2.2. **From the analytic equations to geometric reflection.** Associate to a the period-one Fourier–Bessel series

$$F_a(x+iy) = \sqrt{y}\sum_{n\geq 1} a_n K_{i\nu}(2\pi n y)\big(e^{2\pi i n x} + (-1)^\sigma e^{-2\pi i n x}\big). \tag{2.3}$$

For fixed ν, polynomial coefficient growth and exponential Bessel decay give locally uniform convergence with all derivatives on $y > 0$. Thus F_a is smooth, has no constant Fourier term, has reflection parity $(-1)^\sigma$, and satisfies $-y^2(\partial_x^2 + \partial_y^2)F_a = \lambda_\nu F_a$. The same estimates show that $y^s F_a(x+iy)$ is bounded on $y \geq b$ for every fixed $s \geq 0$ and $b > 0$. Analytic elliptic regularity applies to this equation on every relatively compact subset of $y > 0$ and makes F_a real analytic.

Put $F_{a,(q)}(z) = \sum_{(u,q)=1} F_a(z + u/q)$. The symmetry of the units under $u \mapsto -u$ shows that this is (2.3) with coefficients $a_n c_q(n)$. The functional equation implies

$$F_{a,(q)}(z) = \chi_q F_{a,(q)}\left(-\frac{1}{\ell_N(q)z}\right), \qquad \chi_q = (-1)^\sigma \epsilon_q. \tag{2.4}$$

Here is the parity-sensitive derivation. For a general conductor L and coefficient sequence b, define its axis datum by

$$H_0(y) = F_b(iy), \qquad H_1(y) = \frac{y}{2\pi i}\partial_x F_b(iy).$$

In either case, the Bessel Mellin integral gives

$$2\int_0^\infty H_\sigma(y) y^{s-1/2}\frac{dy}{y} = \Gamma_{\mathbb{R}}(s+\sigma+i\nu)\Gamma_{\mathbb{R}}(s+\sigma-i\nu)L_b(s)$$

initially in a right half-plane. Indeed, for fixed ν one has $K_{i\nu}(u) \ll_{\nu,a} u^{-a}$ on $0 < u \leq 1$ whenever $a > |\operatorname{Im}\nu|$, and exponential decay for $u \geq 1$. Splitting the series at $n = 1/y$ therefore gives $H_\sigma(y) = O(y^{-A})$ for some coefficient-dependent A. This justifies the initial Mellin integral and termwise integration; no common exponent is needed. Inversion uses an absolutely convergent Dirichlet series on a right vertical line and the inverse Mellin transform of its gamma factors. Stirling's formula gives exponential vertical decay on that line; the functional equation gives it on the reflected line. The Phragmén–Lindelöf theorem for an entire function of finite order supplies the intervening strip bounds, with any exponential decay rate strictly below the gamma-factor rate. Contour shifting has no residues and gives

$$H_\sigma(y) = \epsilon_q H_\sigma(1/(Ly)) \qquad (y > 0).$$

For $\sigma = 0$ this is the required equality of values on $x = 0$; the normal derivatives vanish there by even parity. For $\sigma = 1$ both values vanish, while the identity reads

$$\partial_x F_b(iy) = \frac{\epsilon_q}{Ly^2}\partial_x F_b(i/(Ly)).$$

The derivative of $z \mapsto -1/(Lz)$ in the horizontal direction at iy is $-1/(Ly^2)$, explaining the minus sign in χ_q for odd parity. The difference of the two sides of (2.4) therefore has zero value and normal derivative on the imaginary axis and satisfies the same Laplace equation. The equation

determines all higher normal derivatives from these Cauchy data. Real analyticity gives vanishing in a neighborhood of the axis, and connectedness then gives vanishing throughout $\mathbb{H}$. This argument does not presume modularity.

2.3. **A fixed certificate localizes the global norm.** The arithmetic spanning theorem in *Towards a twistless converse theorem* [1] supplies, for every admissible N, a positive integer Q and rational coefficients $A_{j;q,u}$ such that

$$e_j = \sum_{q,u} A_{j;q,u} R_{q,u}, \qquad R_{q,u} = \sum_{\substack{a \bmod q \\ (a,q)=1}} e_{u+aQ/q}, \quad 0 \le j < Q, \tag{2.5}$$

where indices are taken modulo Q, $q \mid Q$, and every used row satisfies the strict inequalities

$$-\frac{Q}{\sqrt{\ell_N(q)}} < u < \frac{Q}{\sqrt{\ell_N(q)}} - 1.$$

This is the sole arithmetic input to the following analytic argument; the spanning theorem and its proof are independent of the spectral parameter. We spell out every constant needed from the certificate. Choose $0 < \theta < 1$ with

$$\frac{\sqrt{\ell_N(q)}}{Q} \max(|u|, |u+1|) \le \theta \quad \text{for each used row,}$$

and put

$$C_0 = \max_j \sum_{q,u} |A_{j;q,u}| \varphi(q), \quad L_* = \max_{A_{j;q,u} \neq 0} \ell_N(q), \quad y_0 = \sqrt{\frac{1-\theta^2}{2L_*}}, \quad \kappa = \frac{2}{1+\theta^2} > 1.$$

Summing coordinates in (2.5) shows $C_0 \ge 1$. Fix an integer $s \ge 0$ such that $C_0 \kappa^{-s} \le 1$. All these choices depend on N alone.

For $z = x + iy$ with $x \in [j/Q, (j+1)/Q]$, set $\beta = z + (u-j)/Q$. Applying the certificate and then (2.4) gives the exact reconstruction

$$F_a(z) = \sum_{q,u} A_{j;q,u} \chi_q \sum_{\substack{a' \bmod q \\ (a',q)=1}} F_a\left(-\frac{1}{\ell_N(q)\beta} + \frac{a'}{q}\right). \tag{2.6}$$

If $0 < y \le y_0$, every resulting height y' satisfies

$$y' = \frac{y}{\ell_N(q)|\beta|^2} \ge \kappa y, \qquad \ell_N(q)|\beta|^2 \le \theta^2 + L_* y_0^2 = \kappa^{-1}.$$

Consequently $y^s |F_a(z)|$ is at most the maximum of $(y')^s |F_a(z')|$ over the finitely many points on the right. Iterate a maximizing branch until its height exceeds y_0. The heights increase by at least κ at each step, so the process terminates. The upper-height Bessel decay already proved yields

$$M := \sup_{x \in \mathbb{R},\ y>0} y^s |F_a(x+iy)| = \sup_{x \in \mathbb{R},\ y \ge y_0} y^s |F_a(x+iy)| < \infty. \tag{2.7}$$

In particular no globally finite norm was assumed before this iteration.

2.4. **A uniform Bessel comparison and an injective coefficient jet.** The only spectral estimate needed is an elementary comparison in the nonoscillating region. Put $W_\nu(v) = \sqrt{v}K_{i\nu}(v)$. Its Bessel equation and large-v asymptotic give

$$W_\nu''(v) = \left(1 - \frac{\lambda_\nu}{v^2}\right) W_\nu(v), \qquad W_\nu(v) \sim \sqrt{\pi/2}\, e^{-v}.$$

For $v \geq \sqrt{\lambda_\nu}$, $W_\nu(v) > 0$: on a putative final positive component ending at infinity, convexity and the derivative limit zero imply that W_ν is decreasing, so that component cannot have a finite zero as its left endpoint. If $\lambda_\nu \leq (1-a^2)v_0^2$, $0 < a < 1$, then on $v \geq v_0$

$$(e^{-av}(W_\nu' + aW_\nu))' \geq 0, \qquad (e^{-v}(W_\nu' + W_\nu))' \leq 0.$$

The standard differentiated Bessel asymptotic also gives $W_\nu'(v) \sim -\sqrt{\pi/2}\, e^{-v}$, so both expressions inside the derivatives tend to zero at infinity. It follows that $-1 \leq W_\nu'/W_\nu \leq -a$, and hence

$$e^{-(v-v_0)} \leq \frac{W_\nu(v)}{W_\nu(v_0)} \leq e^{-a(v-v_0)} \qquad (v \geq v_0). \tag{2.8}$$

The proof covers both spectral branches; it never bounds an oscillating Bessel function below before reaching this region.

Take $\eta = y_0/2$ and $c = \pi\eta/2$. Choose an integer $B \geq 0$ so that

$$\lambda_\nu \leq \tfrac{3}{4}(2\pi n\eta)^2 \ (n > B), \qquad s+1 \leq c(B+1), \qquad \frac{2}{1-e^{-c}} \leq c(B+1). \tag{2.9}$$

One explicit choice is

$$B = \left\lceil \frac{1+|\nu|}{\pi\eta} + \frac{s+1}{c} + \frac{2}{c(1-e^{-c})} \right\rceil, \qquad B \leq C_N(1+|\nu|),$$

since $\lambda_\nu \leq (1+|\nu|)^2$. Suppose $a_1 = \cdots = a_B = 0$. Fourier coefficient extraction on $y = \eta$, justified by absolute convergence, gives

$$|a_n|\sqrt{\eta}\, |K_{i\nu}(2\pi n\eta)| \leq M\eta^{-s}.$$

For $n > B$, apply (2.8) with $a = 1/2$. For $y \geq 2\eta$, writing $v = y/\eta \geq 2$, this yields

$$\begin{aligned} y^s|a_n|\sqrt{y}\, |K_{i\nu}(2\pi ny)| &\leq Mv^s e^{-\pi n\eta(v-1)} \\ &\leq Me^{-1}e^{-cn}. \end{aligned}$$

For the last inequality split the exponential into two equal factors. The first satisfies $v^s e^{-cn(v-1)} \leq e^{-1}$ because $\log v \leq v-1$, $cn \geq s+1$, and $v - 1 \geq 1$; the second is at most e^{-cn}. Also

$$2\sum_{n>B} e^{-cn} = \frac{2e^{-c(B+1)}}{1-e^{-c}} \leq 1$$

by (2.9) and $e^u \geq u$. Summing the positive and negative Fourier modes now gives $y^s|F_a(x+iy)| \leq e^{-1}M$ for $y \geq y_0$. Taking the supremum in (2.7) forces $M \leq e^{-1}M$, hence $M = 0$. Thus $F_a = 0$; extracting its nth coefficient at any height in the positive Bessel region gives $a_n = 0$. We have proved that the coefficient map

$$a \longmapsto (a_1, \ldots, a_B) \quad \text{on } \mathcal{F}_{\nu,\sigma,\epsilon}(N; Q)$$

is injective. In particular its dimension is at most $B = O_N(1+|\nu|)$, uniformly in the phases and the parity.

2.5. **From a dimension bound to a count of Euler products.** Distinct normalized Euler products are linearly independent as coefficient sequences. To see this directly, for every prime p define the linear operator on sequences

$$(T_p a)(n) = a(pn) + \xi_N(p)\mathbf{1}_{p|n}a(n/p).$$

The Euler recurrence gives $T_p a = a_p a$. Two different normalized Euler products differ at some prime, since their prime values determine all prime powers and multiplicativity determines all coefficients. In a shortest nontrivial linear relation among distinct products, apply $T_p - a_p$ for a prime distinguishing two terms. One term is removed and another remains, contradicting minimality. This proves linear independence without requiring the T_p to preserve a fixed sign space.

There are only finitely many sign patterns on the divisors of the fixed integer Q; by (2.2) every Euler product belongs to one of their spaces. The sum of these spaces has dimension at most $2^{d(Q)}B$, where $d(Q)$ is the number of divisors of Q. Every finite family of distinct Euler products is independent in this same finite-dimensional sum. Its size is therefore at most $2^{d(Q)}B$, which proves finiteness of the entire set and gives

$$\#\mathscr{E}_N^{\sigma}(\nu) \le C_N(1+|\nu|) \qquad (\nu \in \mathcal{P}). \tag{2.10}$$

This proves Theorem 1.1. In the complementary range $|\nu| < 1/2$, the bound is $O_N(1)$ for each fixed parameter; it does not assert a bound on a union over different complementary parameters.

3. The lower bound and the genuine new spectrum

3.1. **A quantitative rank-one Lindenstrauss–Venkatesh lower bound.** Throughout this fragment N is fixed. A real spectral parameter is counted once, with $r \ge 0$ and Laplace eigenvalue $1/4 + r^2$. Let $C_M(t,\varepsilon)$ count cuspidal Laplace eigenfunctions on $\Gamma_0(M)\backslash\mathbb{H}$ with $t \le r \le t+\varepsilon$, with multiplicity. Superscripts $+$ and $-$ specify the eigenvalue of the reflection $\iota : z \mapsto -\bar{z}$; their sum is C_M. Write $v_M = \mathrm{vol}(\Gamma_0(M)\backslash\mathbb{H})$.

What is and is not supplied by the sources. Lindenstrauss–Venkatesh [5, Sections 2 and 6] construct operators with cuspidal image and deduce the global Weyl law by a positive pretrace argument. Their Lemma 3 gives an off-diagonal bound for a fixed test kernel, with constants depending on that kernel; its subsequent approximation and limiting argument does not provide a uniform shrinking-window remainder. The argument below is a rank-one quantitative adaptation: we specify the polynomial, the moving spectral test function, and the dependence of the geometric error on both. The stated exponent is proved below, and is not attributed to LV. For an independently established, sharper bound on the genuine total new spectrum, the results of Risager [6, Theorem 4.3 and Proposition 3.1] apply; this separate route is recorded at the end.

Proposition 3.1 (Quantitative LV lower bound)**.** *For fixed M, uniformly for $0 < \varepsilon \le 1$ and $t \ge 3$,*

$$C_M(t,\varepsilon) \ge \frac{v_M}{2\pi}t\varepsilon - O_M\big(t(\log t)^{-1/20}\big), \qquad C_M^{\pm}(t,\varepsilon) \ge \frac{v_M}{4\pi}t\varepsilon - O_M\big(t(\log t)^{-1/20}\big).$$

The exponent $1/20$ is deliberately conservative.

Proof. Fix a prime $p \equiv 1 \pmod{M}$, with $p \nmid M$, and use the normalized unramified Hecke operator T_p. Put

$$D = T_p - 2\cos\big((\log p)\sqrt{\Delta - 1/4}\big).$$

The constant term at every cusp and the operator domain. Normalize each cusp scaling matrix to make its horocycle period one, and write $C_{\mathfrak{a}}f(y) = \int_0^1 f(\sigma_{\mathfrak{a}}(x+iy))\,dx$. For a smooth automorphic function the unramified double-coset calculation gives

$$C_{\mathfrak{a}}(T_p f)(y) = p^{-1/2}C_{\mathfrak{a}_+}f(py) + p^{1/2}C_{\mathfrak{a}_-}f(y/p).$$

To check the two coefficients, use the $p+1$ representatives $z \mapsto pz$ and $z \mapsto (z+j)/p$, $0 \le j < p$, each with coefficient $p^{-1/2}$. For the latter terms the substitution $x \mapsto (x+j)/p$ joins their p intervals into one full horocycle and contributes the factor p. At an arbitrary cusp the same calculation permutes the cusp labels by multiplication by p or p^{-1}; the cusp numerators are taken modulo divisors of M. As $p \equiv 1 \pmod{M}$, both permutations are trivial. For the half-density $C_{\mathfrak{a}} f(y) = y^{1/2} a_{\mathfrak{a}}(\log y)$ this becomes

$$a_{\mathfrak{a}}(u) \longmapsto a_{\mathfrak{a}}(u + \log p) + a_{\mathfrak{a}}(u - \log p).$$

On constant terms $\Delta - 1/4$ becomes $-\partial_u^2$. Thus the one-dimensional wave formula gives the same expression for $2\cos((\log p)\sqrt{\Delta - 1/4})$.

We justify the resulting bounded-operator assertion without treating constant-term extraction as a bounded map on the full L^2 space. The continuous spectrum is a direct integral of Eisenstein representations induced from (χ, χ^{-1}), together with their old vectors at the primes dividing M. Their conductors divide M. At $p \nmid M$ every such old copy has the same spherical Hecke eigenvalue

$$\chi(p) p^{ir} + \chi(p)^{-1} p^{-ir} = 2\cos(r \log p),$$

since $p \equiv 1 \pmod{M}$. Hence D is zero on the entire continuous summand. The residual spectrum with trivial character consists of the constants; there T_p has eigenvalue $p^{1/2} + p^{-1/2} = 2\cosh((\log p)/2)$, also the eigenvalue of the wave expression at $r = i/2$. It is zero on this summand as well. This is the all-cusp intertwining above expressed in the spectral decomposition; no bound for the scattering matrix is used.

Both terms defining D are bounded self-adjoint operators and strongly commute. Indeed, writing $A_p = p^{1/2} + p^{-1/2}$, the Hecke double-coset norm bound is $\|T_p\| \le A_p$ (the positive correspondence has row and column mass A_p); the wave multiplier has absolute value at most A_p on the spectrum $\Delta \ge 0$, including $r \in i[0, 1/2]$. Consequently $\|D\| \le 2A_p$. Since D annihilates the closed noncuspidal summand and is self-adjoint, its image lies in the cuspidal summand. This is the bounded version of the cuspidal-image construction in LV, Sections 2 and 4.

Choose a fixed $B > \|D\|$, for example $B = 4(p+1)/\sqrt{p}$, and set

$$Q_d = I - (I - D^2/B^2)^d \qquad (d \ge 1).$$

The scalar function $1 - (1 - u^2/B^2)^d$ lies in $[0, 1]$ for $|u| \le \|D\|$, so $0 \le Q_d \le I$. Moreover

$$Q_d = \frac{D^2}{B^2} \sum_{j=0}^{d-1} (I - D^2/B^2)^j,$$

which makes its cuspidal image explicit. It commutes with Δ and the reflection ι.

Here and below a "branch" in this polynomial expansion means a rational pullback followed by an archimedean spectral multiplier; the wave operator itself is not a finite sum of rational pullbacks. Put $W = 2\cos((\log p)\sqrt{\Delta - 1/4})$. The binomial formulas expand Q_d into $T_p^j W^l$ with $j + l \le 2d$, and the sum of the absolute scalar coefficients is at most $1 + (1 + 4/B^2)^d$. Each T_p^j expands into $(p+1)^j$ rational pullbacks with coefficient $p^{-j/2}$. The absolute sum of all coefficients is therefore at most C_p^d after increasing a fixed C_p. One may retain W^l as a multiplier, whose real spectral norm is at most 2^l, or expand $(e^{iu \log p} + e^{-iu \log p})^l$ and group the opposite times into cosines. This introduces only another factor 2^{2d} and wave times of absolute value at most $2d \log p$.

The integral representatives $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & j \\ 0 & p \end{smallmatrix}\right)$ have entries at most p. A product of at most $2d$ such matrices has entries at most $(2p)^{2d}$ and determinant at most p^{2d}; dividing out a common factor cannot increase these bounds. Thus branch number, coefficient mass, and rational height are all $\exp(O_p(d))$, and propagation is at most $2d \log p$.

The unramified Plancherel measure of the normalized adjacency coordinate $x \in [-2,2]$ is

$$d\mu_p(x) = \frac{p+1}{2\pi} \frac{\sqrt{4-x^2}}{(\sqrt{p}+p^{-1/2})^2 - x^2}\, dx.$$

Here the normalization is $T_p = p^{-1/2}$ times adjacency on the $(p+1)$-regular tree, and μ_p is a probability measure. One can verify the density and its normalization directly from the root resolvent. If $H(z)$ is the resolvent at the root of a forward p-branch subtree, the Schur complement gives

$$H(z) = \frac{1}{z - H(z)} = \frac{z - \sqrt{z^2-4}}{2}, \qquad G(z) = \frac{1}{z - (1+p^{-1})H(z)}.$$

The square root is chosen to be asymptotic to z at infinity. Taking $-\pi^{-1}\Im G(x+i0)$ gives the displayed density. There are no poles outside $[-2,2]$: there $|H(z)| < 1$ whereas a pole would require $H(z)^2 = p$. At $z = \pm 2$ the denominators of G are $\pm(1-p^{-1})$, so its finite boundary limits exclude endpoint atoms. On the interior of $[-2,2]$ the displayed square-root formula has continuous boundary density, and off this interval it is analytic; hence there is no remaining singular part. Finally $G(z) \sim z^{-1}$ fixes total mass one. Its density is bounded by

$$A_p' = \frac{p+1}{\pi((\sqrt{p}+p^{-1/2})^2 - 4)} < \infty.$$

For every real r, put $a = 2\cos(r \log p) \in [-2,2]$. Our choice of B gives $0 \le (x-a)^2/B^2 < 1$ for $x \in [-2,2]$. The inequality $1 - u \le e^{-u}$ therefore yields

$$\begin{aligned} 0 \le 1 - &\int_{-2}^{2} \left[1 - \left(1 - \frac{(x-a)^2}{B^2}\right)^d\right] d\mu_p(x) \\ &= \int_{-2}^{2} \left(1 - \frac{(x-a)^2}{B^2}\right)^d d\mu_p(x) \\ &\le A_p' \int_{\mathbb{R}} e^{-d(x-a)^2/B^2}\, dx = A_p' B\sqrt{\pi}\, d^{-1/2}. \end{aligned} \tag{3.1}$$

The estimate is uniform in a, including $a = \pm 2$. The integral is exactly the identity coefficient of the finite-place Hecke polynomial: the coefficient of the identity in T_p^j is the normalized number of length-j paths returning to the root, hence $\int x^j\, d\mu_p(x)$. At unramified p the double-coset multiplication on $\Gamma_0(M)$ has this same tree rule. In the geometric expansion, rational branches lying in $\Gamma_0(M)$ combine with the inverse lattice translate to give precisely these identity terms. The other terms have nonidentity real displacement.

We first record an elementary crude local upper bound, needed only to control the tails of a smooth window:

$$C_M(s,1) \ll_M (s+1)\log(s+2). \tag{3.2}$$

Here is a proof that uses neither a local Weyl law nor a shrinking window asymptotic. Fix an even nonnegative bandlimited Schwartz kernel of integral one. Convolve it with the indicator of $[s-1, s+2] \cup [-s-2, -s+1]$, divide by a fixed positive lower bound on $[s, s+1]$, and square. Denote the resulting even multiplier by h_s. It is at least one on $[s, s+1]$, nonnegative on the real spectral axis, and nonnegative on the imaginary axis because its unsquared factor is even and real entire. Its propagation radius is bounded by a fixed constant R_*, independent of s, and

$$\int_{\mathbb{R}} |h_s(r)|\, |r| \tanh(\pi|r|)\, dr \ll s+1.$$

The spherical inversion formula, with $|\varphi_r(u)| \le 1$ for real r, bounds its universal-cover kernel by $C(s+1)$.

On the compact part of the surface the number of lattice elements moving a point by at most R_* is uniformly bounded. In a cusp scaled to period one, the parabolic translations have displacement at most R_* only when $|n| \le 2y\sinh(R_*/2)$; the remaining lattice elements contribute a bounded number below a fixed height and none sufficiently high. Thus the geometric diagonal is at most $C_M(s+1)(1+y)$. Integrating $dy\,dx/y^2$ up to height H at each cusp gives $C_M(s+1)\log(H+2)$.

Every cuspidal eigenfunction f in $[s, s+1]$ has zero horocycle average. The Fourier-series Poincare inequality on a period-one horocycle and the energy identity imply

$$\begin{aligned}\int_{y>H} |f|^2\,d\mu &\le \frac{1}{4\pi^2H^2}\int_{y>H} |\partial_x f|^2\,dx\,dy\\ &\le \frac{1/4+(s+1)^2}{4\pi^2H^2}\|f\|_2^2.\end{aligned}$$

Sum over the finitely many disjoint cusp ends. With $H = C_M(s+1)$ and C_M sufficiently large, every normalized such f has at least half its mass on the truncated surface. In the positive pretrace formula the continuous and residual terms are nonnegative, so they can be discarded for this upper bound. Integrating on the truncated surface gives $\frac{1}{2}C_M(s,1) \le C'_M(s+1)\log(H+2)$, proving (3.2). The finitely many small s are absorbed by increasing the constant.

Here are quantitative geometric details for the moving test function. They also cover the parity insertion. Fix a nonnegative even Schwartz function K of integral one whose Fourier transform is compactly supported. Such a function is obtained by squaring the Fourier transform of a real even compactly supported smooth function and normalizing. Let

$$R = (\log t)^{1/10}, \qquad b = R^{-1/2}, \qquad K_R(u) = RK(Ru),$$
$$J = [t+b, t+\varepsilon-b] \cup [-t-\varepsilon+b, -t-b], \qquad g = 1_J * K_R, \qquad h = g^2.$$

We may assume $\varepsilon \ge 4b$, since otherwise the proposition follows from nonnegativity after increasing its implied constant. Then $0 \le h(r) \le 1$ on the real axis; g is even and real entire, so $h(ir) = g(ir)^2 \ge 0$ for real r as well. The propagation radius of $h(\sqrt{\Delta - 1/4})$ is $O(R)$, and

$$\frac{1}{4\pi}\int_{\mathbb{R}} h(r)r\tanh(\pi r)\,dr = \frac{t\varepsilon}{2\pi} + O(tb+1). \tag{3.3}$$

To verify the normalization, the two intervals defining J have total length $2(\varepsilon - 2b)$. Translation of their indicator by u changes its L^1 norm by at most $4|u|$, and changes its first absolute moment by at most $C(t+1+|u|)|u|$. Integrating these inequalities against $K_R(u)$ gives

$$\int_{\mathbb{R}} (1+|r|)|g(r) - 1_J(r)|\,dr \ll (t+1)/R + R^{-2}.$$

Because $0 \le g \le 1$, we have $|g^2 - 1_J| \le 2|g - 1_J|$. On the positive interval in J, $r\tanh(\pi r) = r + O(te^{-\pi t})$ for large t. The negative interval contributes the same amount, so

$$\int_J r\tanh(\pi r)\,dr = 2\int_{t+b}^{t+\varepsilon-b} r\,dr + O(1) = 2t\varepsilon + O(tb+1).$$

Since $R^{-1} \le b$, these estimates prove (3.3). They also show $\int_0^\infty h(r)r\,dr \ll t$. Since $h \le g$ and $\int g = |J| \le 2$, Cauchy–Schwarz gives $\int_0^\infty h(r)r^{1/2}\,dr \ll t^{1/2}$, uniformly in the allowed ε.

If r lies outside the two original windows, its distance from J is at least b. Integrating the Schwartz tail of K_R beyond that distance gives, for every fixed A,

$$h(r) \ll_A (1 + R\,\mathrm{dist}(r, J))^{-A}.$$

The portions of $[t-1,t+2]$ outside the window therefore contribute at most $C_M t\log t\,(Rb)^{-A}$ by (3.2). On the unit intervals farther away, summation of $(1+Rj)^{-A}$ gives the same bound after increasing the constant; the range $r>2t+2$ can alternatively be summed as $\sum_{n>2t} n\log(n+2)(1+Rn/2)^{-A}=O(t^{-2})$. Hence, taking for example $A=24$,

$$\sum_{\substack{r_j\geq 0\\ r_j\notin[t,t+\varepsilon]}} h(r_j)\ll_M t\log t\,R^{-A/2}+O(t^{-2})\ll_M tR^{-1/2}. \tag{3.4}$$

For the last inequality use $Rb=R^{1/2}$ and $\log t=R^{10}$, so $10-A/2\leq -1/2$ when $A\geq 21$. The sum in this display is over real cuspidal parameters. There are only finitely many imaginary cuspidal parameters. On their fixed strip $|\Im r|\leq 1/2$, integration by parts in the compact Fourier representation of K_R gives, for every fixed L_1,

$$|g(r)|\leq C_{L_1}e^{CR}(1+t)^{-L_1}.$$

Polynomial factors in R are absorbed in e^{CR}. As $R=o(\log t)$, their total h-contribution is $O(t^{-2})$.

Take $d=\lceil R\rceil$ and truncate every cusp at height $H=R^{1/2}$. The omitted area is $O_M(H^{-1})$. Write $A_0=C_M(R+d+1)$. The archimedean kernel in each expanded term of $Q_d h$ has propagation at most A_0; its rational pullback has height at most e^{A_0}. The number and total absolute coefficient of these terms are at most e^{A_0}. On a finite union of fundamental-domain pieces truncated at H, Cartesian coordinates satisfy $|x|,y,y^{-1}\leq H^{C_M}$ after a fixed change of constants. Here is an explicit height/counting argument for lattice translates. If $z=x+iy$ is in the truncated region, the determinant-one matrix $n_x a_y$ carrying i to z and its inverse have entries bounded by H^{C_M}. For g of positive determinant, the condition $d(z,gz)\leq A_0$ implies that the determinant-one representative of $(n_x a_y)^{-1}g(n_x a_y)$ has norm at most $Ce^{A_0/2}$: use its Cartan decomposition, in which the compact factors are orthogonal and the diagonal factor is $\mathrm{diag}(e^{u/2},e^{-u/2})$, $u\leq A_0$. The same argument applies after a reflection if the determinant is negative. The rational branches have primitive determinants of absolute value at most p^{2d}, and left multiplication by a lattice matrix does not increase this bound. Clearing the scalar normalization consequently bounds every entry by

$$B_0=\exp(C_M A_0)H^{C_M}.$$

There are at most $(2\lfloor B_0\rfloor+1)^4$ integral matrices with this entry bound. For each original branch the map from lattice translates to primitive projective matrices is injective up to the harmless sign. Multiplying by the original branch count and coefficient mass, both $e^{O_M(A_0)}$, proves that the total number and total absolute weight of all relevant terms are bounded by $e^{C_M A_0}H^{C_M}$. All constants here are fixed once M and p have been fixed.

For clarity, the small-displacement exceptional sets can be estimated without an ineffective compactness argument. An orientation-preserving branch with integral matrix $\left(\begin{smallmatrix} a & b'\\ c & d'\end{smallmatrix}\right)$ has small displacement only if

$$|cz^2+(d'-a)z-b'|\ll e^{C_M A_0}H^{C_M}\rho.$$

If $c\neq 0$, its imaginary part $(2cx+d'-a)y$ restricts x to an interval of length at most the right side times H^{C_M}; if $c=0$, either the imaginary part is bounded below or the polynomial is a nonzero constant. A scalar matrix is exactly the identity branch and has already been separated. These assertions are uniform because a nonzero integral coefficient has absolute value at least one. If the threshold exceeds the lower bound for a nonzero constant or imaginary part, the trivial rectangle area bound is already at most the claimed polynomial constant times the threshold. Also $y^{-2}\leq H^{C_M}$ on the truncated region, so passing from Cartesian to hyperbolic area only changes that polynomial constant. For an orientation-reversing branch the corresponding polynomial is

$$c(x^2+y^2)+(d'-a)x-b'+i(d'+a)y.$$

If its imaginary part vanishes identically, the remaining real quadratic has sublevel intervals in x of total length $O(\sqrt{\delta})$ for a threshold δ: complete the square and use $|c| \geq 1$. If $c = 0$, its x coefficient is $-2a$; negative determinant forces $a \neq 0$, so the linear sublevel bound applies. If the imaginary part does not vanish identically, $|d' + a| \geq 1$ gives the preceding threshold argument. It follows in both cases, including the hyperbolic area density, that the set where displacement is less than ρ has area at most $e^{C_M A_0} H^{C_M} \rho^{1/2}$. Summing this over the relevant branches has the same form of bound.

Delete all these sets with $\rho = t^{-1/8}$ and call the resulting region Ω_0. We give the off-diagonal kernel estimate used on this region, including its dependence on A_0 and ρ. For real r the spherical function has angular representation

$$\varphi_r(u) = \frac{1}{2\pi} \int_0^{2\pi} (\cosh u - \sinh u \cos\theta)^{-1/2+ir} \, d\theta.$$

Put $a_u(\theta) = (\cosh u - \sinh u \cos\theta)^{-1/2}$ and $\psi_u(\theta) = \log(\cosh u - \sinh u \cos\theta)$. For $\rho \leq u \leq A_0$,

$$\|a_u\|_\infty + \mathrm{Var}(a_u) \leq C e^{CA_0}, \qquad |\psi_u'(\theta)| = \frac{\sinh u \, |\sin\theta|}{\cosh u - \sinh u \cos\theta}.$$

If $r\rho \geq 16$, remove angular intervals of length $\delta = (r\rho)^{-1/2}$ around 0 and π. Their contribution is at most $Ce^{CA_0}\delta$. On the remaining arcs $|\psi_u'| \geq ce^{-A_0}\rho\delta$. The derivative is monotone after splitting each half-circle at $\cos\theta = \tanh u$, hence on a bounded number of arcs. The first-derivative integration-by-parts bound, followed by partial summation for the amplitude a_u, bounds the remaining integral by $Ce^{CA_0}/(r\rho\delta)$. If $r\rho < 16$, the trivial bound $|\varphi_r(u)| \leq 1$ gives the same conclusion with a larger constant. We have proved

$$|\varphi_r(u)| \leq C e^{CA_0} (r\rho)^{-1/2} \quad (r \geq 1, \ \rho \leq u \leq A_0). \tag{3.5}$$

Thus no implicit stationary-phase constant is allowed to depend uncontrolledly on the growing propagation radius.

After the polynomial expansion each archimedean multiplier is $h(r)$ times a product of at most $2d$ factors $2\cos(r \log p)$, or its finite cosine expansion. Its absolute value on the real axis is bounded by $2^{2d} h(r)$. Spherical inversion, with Plancherel density $r \tanh(\pi r)/(4\pi)$ on the whole real line, and (3.5) bound its kernel at displacement $\rho \leq u \leq A_0$ by

$$C e^{C_M A_0} \rho^{-1/2} \int_1^\infty h(r) r^{1/2} \, dr + O(t^{-2}) \leq C e^{C_M A_0} t^{1/2} \rho^{-1/2}.$$

The part $0 \leq r \leq 1$ is $O(t^{-2})$ by the real Schwartz tail of h, even after multiplication by $e^{C_M d}$. There is no imaginary-spectrum integral in this universal-cover inversion formula. Finite propagation makes the kernel zero for $u > A_0$.

Multiplying by the previously established branch counts and coefficient mass, integrating over Ω_0, and using $\mathrm{vol}(\Omega_0) \leq v_M$ now gives

$$\left| \int_{\Omega_0} K_{Q_d h}(z,z) \, d\mu(z) - \mathrm{vol}(\Omega_0) \, k_{Q_d h}(1) \right| \ll_M e^{C_M(R+d)} H^{C_M} t^{9/16}. \tag{3.6}$$

Here $t^{9/16} = t^{1/2}\rho^{-1/2}$.

For $(I \pm \iota) Q_d h/2$, the identity contribution is divided by two, and all reflection terms obey the same error bound. The removed small-displacement area costs, by (3.3), at most $e^{C_M(R+d)} H^{C_M} t^{15/16}$ in the main term.

For completeness, the final spectral comparison uses positivity only after the polynomial has been reassembled. Choose an orthonormal joint eigenbasis f_j of the cuspidal Laplace and Hecke operators, and restrict to a parity when needed. If q_j is the eigenvalue of Q_d, then $0 \leq q_j \leq 1$. The smoothing

supplied by h makes the diagonal expansion locally absolutely convergent, and Tonelli gives

$$\int_{\Omega_0} K_{Q_d h}(z,z)\,d\mu(z) = \sum_j q_j h(r_j) \int_{\Omega_0} |f_j(z)|^2\,d\mu(z) \le \sum_j q_j h(r_j).$$

Both real and imaginary summands are nonnegative. As $0 \le h \le 1$ on the real axis, the last sum is at most $C_M(t,\varepsilon)$ plus the tails estimated in (3.4), and an $O(t^{-2})$ imaginary term. The same inequalities hold in either parity subspace.

Writing $q_0(r)$ for the identity coefficient in (3.1), the universal identity kernel is

$$k_{Q_d h}(1) = \frac{1}{4\pi} \int_{\mathbb{R}} q_0(r) h(r) r \tanh(\pi r)\,dr.$$

It is between zero and Ct, and differs from the mass in (3.3) by at most $C_p t d^{-1/2}$. This also justifies charging the deleted volume by its area rather than by a pointwise bound for the full periodized kernel. The cusp truncation loses area $O_M(H^{-1})$, and the exceptional sets lose area at most $e^{C_M(R+d)} H^{C_M} \rho^{1/2}$. Combining all estimates, with $\kappa = 1$ for both parities and $\kappa = 1/2$ for one parity, yields

$$C_M^{(\kappa)}(t,\varepsilon) \ge \kappa \frac{v_M}{2\pi} t\varepsilon - O_M\left(t(b + H^{-1} + d^{-1/2}) + e^{C_M(R+d)} H^{C_M} t^{15/16} \right).$$

The smaller off-diagonal error $t^{9/16}$ has been absorbed into $t^{15/16} = t\rho^{1/2}$.

To verify every parameter inequality, recall

$$R = (\log t)^{1/10}, \quad b = R^{-1/2}, \quad d = \lceil R \rceil, \quad H = R^{1/2}.$$

For large t, $R \le d \le 2R$, so the first error is at most $C_M t R^{-1/2} = C_M t (\log t)^{-1/20}$. Furthermore

$$\log\bigl(e^{C_M(R+d)} H^{C_M}\bigr) \le C'_M (\log t)^{1/10} + C'_M \log\log t \le \tfrac{1}{32} \log t$$

for sufficiently large t, with threshold depending only on M and the fixed p. The geometric error is consequently at most $C_M t^{31/32}$, which is $o(t(\log t)^{-1/20})$. The discarded case $\varepsilon < 4b$ satisfies $t\varepsilon \le 4t(\log t)^{-1/20}$ and follows from nonnegativity. Finally a fixed bounded range of $t \ge 3$ is absorbed by enlarging the constant. This proves the asserted uniform statement for all $0 < \varepsilon \le 1$.

□

3.2. **A proved sharper benchmark and passage to exact level.** For the total genuine spectrum there is a stronger existing theorem. Risager [6, Theorem 4.3] gives for each fixed M

$$C_M([0,T]) = \frac{v_M}{4\pi} T^2 + a_M T \log T + b_M T + O_M(T/\log T). \tag{3.7}$$

Changing between T and $\sqrt{1/4 + T^2}$ changes only the stated error, and constant or residual eigenfunctions change the count by $O_M(1)$. The cited paper states the formula in the latter variable. This is the cited Risager input, not a conclusion of the preceding LV construction. To see exactly what differencing gives, put $F(T) = v_M T^2/(4\pi) + a_M T \log T + b_M T$. Uniformly for $0 < \varepsilon \le 1$ and $t \ge 3$,

$$F(t+\varepsilon) - F(t) = \frac{v_M}{2\pi} t\varepsilon + O_M(\varepsilon \log t + \varepsilon^2),$$

by the identity $(t+\varepsilon)^2 - t^2 = 2t\varepsilon + \varepsilon^2$ and the mean value theorem for $T \log T$. The two endpoint remainders each have size $O_M(t/\log t)$; no cancellation between them is asserted. Consequently,

$$C_M(t,\varepsilon) = \frac{v_M}{2\pi} t\varepsilon + O_M(t/\log t + \varepsilon \log t + \varepsilon^2 + 1). \tag{3.8}$$

Endpoint multiplicities are also $O_M(t/\log t)$, by applying (3.7) on intervals of arbitrarily small length, so the choice of closed or half-open windows does not affect this.

Combining the total upper bound (3.8) with the two parity lower bounds proves

$$C_M^{\pm}(t,\varepsilon) = \frac{v_M}{4\pi} t\varepsilon + O_M(t(\log t)^{-1/20}).$$

Indeed, to get the upper estimate for one parity, subtract the LV lower estimate for the opposite parity from the total Risager upper estimate; use $C_M = C_M^+ + C_M^-$. All additional terms $t/\log t$, $\varepsilon \log t$, ε^2 and 1 are $O(t(\log t)^{-1/20})$ for $0 < \varepsilon \le 1$. Together with the two LV lower estimates this proves the display. Thus the parity conclusion uses both the quantitative LV argument and the cited total-spectrum theorem; it is not being inferred from an unquantified equivariant Weyl law.

For a spectral interval I, the exact oldform decomposition gives

$$C_N(I) = \sum_{d|N} d(N/d) C_d^{\mathrm{new}}(I), \qquad C_N^{\mathrm{new}}(I) = \sum_{d|N} \beta(N/d) C_d(I), \qquad \beta = \mu * \mu.$$

Here $d(n) = \sum_{a|n} 1$ is the divisor function. A newform of exact level $d \mid N$ contributes the old vectors $f(az)$ for $a \mid N/d$, a space of dimension $d(N/d)$ with the same spectral parameter. The Dirichlet inverse of $d = 1 * 1$ is $\mu * \mu$, which proves the second identity from the first. The same identities hold separately for the two parities, because $f(z) \mapsto f(az)$ preserves reflection parity. This is an identity of finite-dimensional eigenspaces, and thus holds for an arbitrary interval before any asymptotic is taken; see [6, Lemma 3.1 and Proposition 3.1]. Writing

$$\nu(N) = \prod_{p^a \| N} \nu_p(a), \qquad \nu_p(1) = p - 1,$$
$$\nu_p(a) = I_p(a) - 2I_p(a-1) + I_p(a-2),$$
$$I_p(0) = 1, \qquad I_p(a) = (p+1)p^{a-1},$$

with the second formula for $a \ge 2$, we obtain the formulas below. For their leading constants, use $v_M = (\pi/3) I(M)$, where I is multiplicative with the prime-power values $I_p(a)$ just specified. Convolution with $\mu * \mu$ gives $(\mu * \mu * I)(N) = \nu(N)$, hence the total coefficient $(\pi/3)\nu(N)/(2\pi) = \nu(N)/6$. A fixed parity divides this coefficient by two. The finitely many remainder terms from divisors of the fixed N are absorbed into $O_N(\cdot)$:

$$C_N^{\mathrm{new}}(t,\varepsilon) = \frac{\nu(N)}{6} t\varepsilon + O_N(t/\log t + \varepsilon \log t + \varepsilon^2 + 1), \tag{3.9}$$

$$C_N^{\mathrm{new},\pm}(t,\varepsilon) = \frac{\nu(N)}{12} t\varepsilon + O_N(t(\log t)^{-1/20}). \tag{3.10}$$

For the admissible levels in the main manuscript, $\nu(N) = \varphi(N)$ for squarefree N, $\nu(4m) = \varphi(m)$, and $\nu(8m) = 3\varphi(m)$. In particular the total new asymptotic holds whenever $\varepsilon(t) \log t \to \infty$; the parity refinement proved here holds whenever $\varepsilon(t)(\log t)^{1/20} \to \infty$. Both conditions follow from the shrinking-window hypothesis $\varepsilon(t) \log\log\log t \to \infty$.

One can retain the quantitative LV construction as the lower-bound input at the new level itself. Use the positive oldform identity in the form

$$C_N^{\mathrm{new}}(t,\varepsilon) = C_N(t,\varepsilon) - \sum_{d|N,\ d<N} d(N/d) C_d^{\mathrm{new}}(t,\varepsilon).$$

Insert Proposition 3.1 for the first term and the genuine new-spectrum upper estimates at proper divisors for the subtracted terms. There are only finitely many such divisors. This proves

$$C_N^{\mathrm{new}}(t,\varepsilon) \ge \frac{\nu(N)}{6} t\varepsilon - O_N(t(\log t)^{-1/20}),$$

and the identical argument at fixed parity gives coefficient $\nu(N)/12$. Thus the lower bound for Euler products below uses the LV cuspidal-image construction, with known upper bounds only to remove the old spectrum; it does not attempt to difference an unquantified global Weyl law.

Connection and scope. Every normalized genuine Maass newform produces an admissible Euler-product solution, and distinct normalized newforms give distinct coefficient sequences by multiplicity one. Thus (3.9) is a rigorous lower bound for the total solution count, summed over parity and ramified/Fricke signs. If the gamma parity is fixed, use (3.10) instead. This argument supplies neither an upper bound for nonmodular solutions at varying r nor their orthogonality in a common Hilbert space; that is a separate bridge problem.

For a fully specified ramified/Fricke sector the predicted fixed-parity coefficient is $c_N(\boldsymbol{\delta})$ from the main manuscript, and without fixing parity it is $2c_N(\boldsymbol{\delta})$. This follows at the level of the candidate Plancherel coefficient from $\tau_\Gamma(P) = 4\pi c_N(\boldsymbol{\delta})$. The argument above proves the total and parity counts and does not silently promote this trace computation to a proved sector Weyl law. A sector version requires either a quantitative equivariant trace formula or finite approximations of the local-new projection with a controlled error on the actual automorphic old spectrum. The regular-tree gap alone controls only the regular module and is insufficient for that last assertion.

4. The physical target and the unit-width trace comparison

Exact nonzero-channel coordinates and Hilbert–Schmidt stability. Let $\mathcal{T} = L^2((\mathbb{R}/\mathbb{Z}) \times [\varepsilon_0, \infty), y^{-2}dx\,dy)$ be the raw terminal Hilbert space, with fixed terminal height $\varepsilon_0 > 0$. Let Π_0 average over x at each height, and put $\Pi = I - \Pi_0$. Suppose R is bounded, is terminal inclusion on the terminal strip, and put $G = R^*R$, so $I \le G \le CI$. All inverses on $\Pi\mathcal{T}$ below are taken in that Hilbert space.

Lemma 4.1 (Compressed Gram coordinates)**.** *Set*

$$K = \Pi G \Pi\big|_{\Pi\mathcal{T}}, \qquad U_! = R\Pi K^{-1/2}, \qquad J = G^{1/2}\Pi K^{-1/2}.$$

Then $U_!$ and J are isometries, $U_! = UJ$ for $U = RG^{-1/2}$, and

$$JJ^* = P_{G^{1/2}\Pi\mathcal{T}} = G^{1/2}\Pi K^{-1}\Pi G^{1/2}.$$

If an actual solution has raw terminal datum $h \in \Pi\mathcal{T}$, then its compressed normalized datum $g_! = K^{1/2}h$ satisfies $U_!g_! = Rh$ and $Jg_! = G^{1/2}h$.

Proof. The identities $J^*J = K^{-1/2}\Pi G\Pi K^{-1/2} = I$ and $U_!^*U_! = I$ are immediate. Since $G^{1/2}$ is invertible, its image of the closed subspace $\Pi\mathcal{T}$ is closed; it is exactly the range of J. The remaining assertions follow by multiplication. □

The original known-residual polynomial can be constructed directly using $U_!$ in place of U. Its scalar norm bound remains valid, because $U_!$ is an isometry. It fixes every actual solution's datum $g_!$ exactly. There is no need to estimate a separate difference of range projections $P - P_Y$.

The regular model of the channel projection. For $j \ge 1$ let

$$E_j = 2^{-j}\sum_{a=0}^{2^j-1} T_{a/2^j}.$$

On the physical cylinder E_j is the orthogonal projection onto Fourier frequencies divisible by 2^j, so $E_j \downarrow \Pi_0$ strongly. On each regular translation-orbit fiber it is the projection onto vectors invariant under the finite subgroup $2^{-j}\mathbb{Z}/\mathbb{Z}$. The union $\mathbb{Z}[1/2]/\mathbb{Z}$ has infinite free orbits on that fiber. An invariant square-summable vector on any such orbit is zero. Hence $E_j \downarrow 0$ strongly in the regular representation: the regular realization of Π is I. Equivalently, in the finite terminal corner $\tau(E_j) = 2^{-j}\tau(I)$, since every nonidentity translation has zero regular trace; normality and

faithfulness give the same result. All these averages are integral at every prime $q \neq 2$. Thus the compressed normalization has exactly the previous regular model, and excluding the single prime 2 preserves all freshness and density-one assertions. This argument is applied fiberwise away from the null stabilizer locus, as in the regular orbit model.

A fixed-propagation cylinder majorant. Let Δ be the cylinder Laplacian and put

$$M_t = \left(1 + \left(\frac{\Delta - (1/4 + t^2)}{t}\right)^2\right)^{-4}, \qquad t \geq 2.$$

Choose an even nonnegative entire function j of fixed exponential type satisfying

$$c(1 + r^2)^{-4} \leq j(r) \leq C(1 + r^2)^{-4} \quad (r \in \mathbb{R}).$$

For example sum $\operatorname{sinc}(r)^8$ and its translates by $\pm\pi/2$ with symmetric positive coefficients. Their zeros do not coincide. Comparison for large $|r|$ and then on a fixed compact interval proves the two-sided bound. Let J_t have multiplier $j(r - t) + j(r + t)$. The cylinder spectrum has $r \in \mathbb{R}$, and the scalar inequality

$$\left(1 + \left(\frac{r^2 - t^2}{t}\right)^2\right)^{-4} \leq C\big(j(r - t) + j(r + t)\big)$$

gives $0 \leq M_t \leq CJ_t$. The operator J_t has a fixed propagation radius R_0, independent of t.

Spherical inversion gives for its hyperbolic kernel

$$|k_{J_t}(u)| \leq Ct, \qquad |k_{J_t}(u)| \leq C\sqrt{t}\, u^{-1/2} \quad (0 < u \leq R_0). \tag{4.1}$$

Indeed $|\varphi_r(u)| \leq C \min(1, (|r|u)^{-1/2})$ on the fixed propagation ball, by the elementary angular stationary-phase estimate, and $\int_0^\infty (j(r-t)+j(r+t))r\, dr \ll t$ and $\int_0^\infty (j(r-t)+j(r+t))\sqrt{r}\, dr \ll \sqrt{t}$. The polynomial decay of j suffices for all these inversions. At most $C(1 + y)$ integer translates of $z = x + iy$ are within this fixed propagation radius. Periodization and positive operator order consequently give the diagonal bound

$$M_t(z, z) \leq Ct(1 + y). \tag{4.2}$$

Kernel diagonal comparisons can equivalently be obtained by testing positive operator order on smooth approximate delta functions.

Lemma 4.2 (Exact nonzero-channel cusp trace)**.** *For every $H > 0$,*

$$\operatorname{tr}\big(M_t \Pi \mathbf{1}_{y>H}\big) \leq \frac{1}{2\pi H} \int_0^\infty m_t(r) r \tanh(\pi r)\, dr \ll \frac{t}{H}, \quad m_t(r) = \left(1 + \left(\frac{r^2 - t^2}{t}\right)^2\right)^{-4}. \tag{4.3}$$

The trace means that of the positive localization $\mathbf{1}_{y>H} \Pi M_t \Pi \mathbf{1}_{y>H}$.

Proof. In Fourier mode $n \neq 0$, put $u = \log y$ and write the function as $y^{1/2} f(u) e^{2\pi i n x}$. This is a unitary transformation to $L^2(du)$, and the Laplacian becomes

$$-\partial_u^2 + 1/4 + (2\pi n)^2 e^{2u}.$$

The Kontorovich–Lebedev transform diagonalizes this operator with generalized eigenfunctions

$$\psi_r(x) = \frac{\sqrt{2r \sinh(\pi r)}}{\pi} K_{ir}(x), \qquad x = 2\pi|n|y, \quad r > 0,$$

and measure dr. This normalization is the transform/inversion pair in NIST DLMF [3, equations 10.43.30–31]. More explicitly, the transform is $\widehat{f}(r) = \int_0^\infty f(x)\psi_r(x)\, dx/x$, with inverse $f(x) = \int_0^\infty \widehat{f}(r)\psi_r(x)\, dr$. The cited inversion formula first applies to smooth compactly supported functions. Applying it to their inner products gives Parseval, and density gives the unitary spectral resolution on $L^2(dx/x)$. The Bessel differential equation then gives the displayed Laplacian. As a check on the

normalization, each of the two small-x oscillatory terms of $\psi_r(e^u)$ has coefficient of modulus $1/\sqrt{2\pi}$; this check alone is not a completeness argument.

For clarity the Bessel integral needed here can be evaluated directly. The Schlaefli representation [3, equation 10.32.10] gives

$$K_{ir}(x)K_{-ir}(x) = \frac{1}{4}\int_0^\infty \int_0^\infty u^{-ir-1}v^{ir-1}\exp\left(-u-v-\frac{x^2(u+v)}{4uv}\right)du\,dv.$$

For real $s>0$, the integral of x^{s-1} times the last exponential factor is

$$\int_0^\infty x^{s-1}e^{-x^2(u+v)/(4uv)}dx = 2^{s-1}\Gamma(s/2)\left(\frac{uv}{u+v}\right)^{s/2}.$$

After multiplication by $u^{-ir-1}v^{ir-1}/4$, change variables to $w=u+v$, $z=u/(u+v)$, with Jacobian w. The two remaining integrals are

$$2^{s-3}\Gamma(s/2)\int_0^\infty e^{-w}w^{s/2-1}dw\int_0^1 z^{s/2-ir-1}(1-z)^{s/2+ir-1}dz.$$

Both are absolutely convergent: removing the factors of modulus one reduces the last integral to $B(s/2,s/2)$, finite because $s>0$. This also supplies an absolute majorant for the initial threefold integral and justifies Fubini before the substitutions. Since $K_{-ir}=K_{ir}$ is real on the positive axis, evaluation of the gamma and beta integrals yields

$$\int_0^\infty x^{s-1}K_{ir}(x)^2\,dx = \frac{2^{s-3}}{\Gamma(s)}\Gamma(s/2)^2\Gamma(s/2+ir)\Gamma(s/2-ir).$$

At $s=1$, use $\Gamma(1/2)^2=\pi$ and $\Gamma(1/2+ir)\Gamma(1/2-ir)=\pi/\cosh(\pi r)$ to obtain

$$\int_0^\infty K_{ir}(x)^2\,dx = \frac{\pi^2}{4\cosh(\pi r)}, \qquad \int_0^\infty |\psi_r(x)|^2\,dx = \frac{r\tanh(\pi r)}{2}. \tag{4.4}$$

The unitary spectral resolution shows that the positive localized trace in mode n is

$$\int_0^\infty m_t(r)\int_H^\infty |\psi_r(2\pi|n|y)|^2\frac{dy}{y}\,dr.$$

This identity can first be read as an extended nonnegative trace; the following bound proves its finiteness. Tonelli's theorem allows all sums and integrals to be interchanged. For positive Fourier modes,

$$\begin{aligned}\sum_{n\ge1}\int_H^\infty |\psi_r(2\pi ny)|^2\frac{dy}{y} &= \int_0^\infty \left\lfloor\frac{x}{2\pi H}\right\rfloor|\psi_r(x)|^2\frac{dx}{x}\\ &\le \frac{1}{2\pi H}\int_0^\infty |\psi_r(x)|^2\,dx = \frac{r\tanh(\pi r)}{4\pi H}.\end{aligned}$$

Negative modes give an identical contribution. Integrating against $m_t(r)\,dr$ proves the stated inequality. Finally, the substitution $v=(r^2-t^2)/t$ gives the explicit bound

$$\int_0^\infty m_t(r)r\,dr = \frac{t}{2}\int_{-t}^\infty (1+v^2)^{-4}dv \le \frac{5\pi t}{32}.$$

Thus the trace is at most $5t/(64H)$, and in particular at most $t/(8H)$. No estimate for the full, unprojected infinite cusp trace has been used. □

Low reconstruction tails and normalization errors. Work in the unweighted realization of the weighted ambient space. Choose the reconstruction weight sufficiently strong that, for some fixed $a > 1/2$, its low part factors as

$$\mathbf{1}_{y<\varepsilon_0} R = \mathbf{1}_{y<\varepsilon_0} b(y) R_0, \qquad |b(y)| \le C y^a, \qquad \|R_0\| \le C.$$

Such a factorization is precisely boundedness of the same reconstruction in a weaker fixed polynomial weight. Equation (4.2) gives

$$\operatorname{tr}(M_t \mathbf{1}_{y<Y} R R^* \mathbf{1}_{y<Y}) \ll t Y^{2a-1} \quad (0 < Y < \min(\varepsilon_0, 1)). \tag{4.5}$$

Indeed the operator between the two outer height cutoffs is bounded above by $C\mathbf{1}_{y<Y} y^{2a}$, and the diagonal integral is $O(t\int_0^Y y^{2a-2}dy)$. Split $R\Pi$ into its low part and its terminal inclusion $\mathbf{1}_{y\ge\varepsilon_0}\Pi$. The inequality $(A+B)(A+B)^* \le 2AA^* + 2BB^*$ and Lemma 4.2 then give

$$\operatorname{tr}(M_t R \Pi R^*) \ll_{\varepsilon_0} t. \tag{4.6}$$

Let $\mathcal{W} = w h_{t,R}(\sqrt{\Delta - 1/4}) w^{-1}$ satisfy the weighted unit-majorant estimate $\mathcal{W}^*\mathcal{W} \le C e^{CR} M_t$. Set $R_Y = \mathbf{1}_{y\ge Y} R$, $K_Y = \Pi R_Y^* R_Y \Pi$, and $U_{!,Y} = R_Y \Pi K_Y^{-1/2}$. Then

$$\|K - K_Y\| \ll Y^{2a}, \qquad \|K^{-1/2} - K_Y^{-1/2}\| \ll Y^{2a},$$

Indeed $K - K_Y = \Pi R^* \mathbf{1}_{y<Y} R \Pi$, whose norm is $O(Y^{2a})$ by the low-part factorization. Both compressed Grams are at least I, since $Y < \varepsilon_0$ retains the terminal inclusion. For positive operators $A, B \ge I$, the resolvent identity and the norm-convergent integral

$$A^{-1/2} = \frac{1}{\pi}\int_0^\infty s^{-1/2}(A + sI)^{-1} ds$$

give

$$\|A^{-1/2} - B^{-1/2}\| \le \frac{\|A - B\|}{\pi}\int_0^\infty \frac{s^{-1/2}}{(1+s)^2} ds = \tfrac{1}{2}\|A - B\|.$$

This proves the second bound without assuming that A and B commute. The reconstruction difference is exactly

$$U_! - U_{!,Y} = (R - R_Y)\Pi K_Y^{-1/2} + R\Pi(K^{-1/2} - K_Y^{-1/2}).$$

For every bounded source operator D on $\Pi\mathcal{T}$, the Hilbert–Schmidt ideal inequality and positive operator order give

$$\|\mathcal{W} R \Pi D\|_{\mathrm{HS}}^2 \le C e^{CR} \|D\|^2 \operatorname{tr}(M_t R \Pi R^*).$$

The same statement for $R - R_Y$ is bounded by the low-tail trace in (4.5). Apply these two inequalities to the two summands of the exact difference, using $\|K_Y^{-1/2}\| \le 1$ and $\|K^{-1/2} - K_Y^{-1/2}\| = O(Y^{2a})$. The inequality $\|S+T\|_{\mathrm{HS}}^2 \le 2\|S\|_{\mathrm{HS}}^2 + 2\|T\|_{\mathrm{HS}}^2$ then proves

$$\|\mathcal{W} U_!\|_{\mathrm{HS}}^2 \ll e^{CR} t, \qquad \|\mathcal{W}(U_! - U_{!,Y})\|_{\mathrm{HS}}^2 \ll e^{CR} t \left(Y^{2a-1} + Y^{4a}\right). \tag{4.7}$$

Uniform compressed-Gram polynomial errors ζ contribute at most $C e^{CR} t \zeta^2$ by the same argument. These statements prove genuine Hilbert–Schmidt compactness while retaining the nonzero source channel exactly. A naked norm error multiplying the full terminal identity would not justify them.

Lemma 4.3 (Internal zero-channel factors)**.** *Let $A = \sum_i a_i U_{g_i}$ be a finite sum of partial rational isometries between folded cylinder pieces, with all source and image heights in $[Y, H]$, where $0 < Y \le 1 \le H$. The coefficients may be arbitrary bounded measurable functions. Set $\Lambda = \sum_i \|a_i\|_\infty$. Then*

$$\|M_t^{1/2} A \Pi_0\|_{\mathrm{HS}}^2 \le C\Lambda^2 (1 + H) Y^{-1/2} t^{1/2}. \tag{4.8}$$

Here a source or image height cutoff may be absorbed into a_i. For any bounded B, the right side multiplied by $\|B\|^2$ also bounds $\|M_t^{1/2} A \Pi_0 B\|_{\mathrm{HS}}^2$.

Proof. Positive operator order lets us replace M_t by CJ_t. It suffices to treat one branch aU_g with $\|a\|_\infty \le 1$; the Hilbert–Schmidt triangle inequality then gives the stated coefficient cost. We justify the kernel calculation while proving that this norm is finite. The continuous positive cylinder kernel $J_t(z,z')$ has a Hilbert-space factorization $J_t(z,z') = \langle q_{z'}, q_z\rangle$, where q_z is the kernel column of $J_t^{1/2}$ and $\|q_z\|^2 = J_t(z,z) < \infty$. One can obtain these columns first on compact height intervals from the spectral resolution of J_t, or equivalently by the reproducing-kernel construction for its positive kernel. This construction identifies the resulting integral operator with $J_t^{1/2}$ on compactly supported test functions.

Identify the source zero channel with $L^2(dy/y^2)$. Changing variables by the isometry g shows that the kernel column of $J_t^{1/2}(aU_g)\Pi_0$ at height y is a horizontal integral of the vectors $q_{g^{-1}(x+iy)}$, with the transformed coefficient and domain mask. Hyperbolic measure is invariant, so there is no additional Jacobian factor. These are Bochner integrals on the retained compact-height pieces. Their squared norms are the double horizontal integrals of J_t with the two coefficient factors. Integration in dy/y^2 gives the squared Hilbert–Schmidt norm if finite, and gives its extended value otherwise. Equivalently, the zero-channel kernel relative to hyperbolic measure is $y^2\delta(y-y')$.

Write $\chi_y(x)$ for the transformed domain mask; it takes values in $[0,1]$. Since $|a| \le 1$, the squared norm is bounded by

$$\int_Y^H \frac{dy}{y^2}\int_0^1\int_0^1 \chi_y(x)\chi_y(x')\sum_n \left|k_{J_t}\big(d(g^{-1}(x+iy), T_n g^{-1}(x'+iy))\big)\right| dx\,dx'.$$

On the support of these masks, source and image folds have real coordinates in $[0,1)$ and heights at most H. If a summand is nonzero, fixed propagation and the distance formula imply $|\Re(g^{-1}(x+iy)) - \Re(g^{-1}(x'+iy)) - n| \le C_{R_0}H$. Consequently every contributing integer lies in a fixed interval $|n| \le C_{R_0}(1+H)$, independently of x, x', y. Only after restricting to this finite interval do we discard the masks. Discarding them before restricting the periodization sum would not justify a uniform count of its terms. For each such n put $\eta_n = gT_ng^{-1}$. Isometry invariance turns the distance in the last display into

$$u = d(x+iy, \eta_n(x'+iy)).$$

Fix x' and y, and write $\eta_n(x'+iy) = a_0 + ib_0$. Whenever $u \le R_0$, one has $e^{-R_0}y \le b_0 \le e^{R_0}y$ and the hyperbolic distance formula gives

$$u \ge c_{R_0}|x-a_0|/y.$$

Equation (4.1) therefore gives

$$\int_0^1 |k_{J_t}(u)|\,dx \le C\sqrt{ty}\int_0^1 |x-a_0|^{-1/2}\,dx \le C\sqrt{ty}.$$

The singularity is integrable; no smoothness or derivative bound on a has been used. Integrating x' and then y contributes at most $C\sqrt{t}\int_Y^H y^{-3/2}dy \le C\sqrt{t}\,Y^{-1/2}$. Summing the $O(1+H)$ translates proves that the Hilbert-valued kernel columns have finite squared integral, and hence proves the single-branch Hilbert–Schmidt estimate. The displayed absolute bound also justifies Fubini in the kernel calculation. In particular the proof never assumes trace class in order to use a trace formula, and measurable domain masks cause no difficulty. The assertion with B is the ideal property of that norm. □

How compressed-Gram words are compared. In a finite polynomial approximation expand each Π as $I - \Pi_0$. Terms containing no Π_0 are the usual rational finite-branch words. In every other term choose the leftmost Π_0 and write the term as $A\Pi_0 B$, where A is a finite branch sum and B is bounded. When the compact terminal block is in use, all source and image heights of A are retained in a common interval $[Y,H]$. Lemma 4.3 controls the term without imposing regularity on

its coefficient functions or expanding the remaining channel factors in B. The weighted wave estimate adds only Ce^{CR}. Thus a total coefficient/geometry cost $\mathcal{C}$ yields squared Hilbert–Schmidt channel error $O(e^{CR}\mathcal{C}t^{1/2})$, and cross terms with an $O(\sqrt{t})$ Hilbert–Schmidt operator are $O(e^{CR}\mathcal{C}t^{3/4})$ after enlarging the cost. These are power savings when the finite recipe has cost $t^{o(1)}$.

The high terminal block of R_Y and its compressed Gram is exactly the identity once its height exceeds all retained reconstruction samples. On that block the source projection is exactly Π. Lemma 4.2 then gives an $O(t/H)$ unit-majorant trace error, with the weighted-wave factor e^{CR} when appropriate. This is compatible with subpolynomial cusp cutoffs. On the direct compact zero channel one additionally has $\operatorname{tr}(M_t\Pi_0\mathbf{1}_{\varepsilon_0\le y\le H}) = O_{\varepsilon_0}(\log(2+H))$. Indeed the half-density change $u = \log y$ makes this channel $-\partial_u^2 + 1/4$ on the full line, whose localized multiplier trace is exactly

$$\frac{\max(0, \log(H/\varepsilon_0))}{2\pi} \int_{\mathbb{R}} m_t(|r|)\, dr.$$

For $r \ge 0$, $m_t(r) \le (1+(r-t)^2)^{-4}$, so the last integral is bounded uniformly for $t \ge 2$. This sharper direct-channel identity is not needed for the internal channel factors.

4.1. **A common Maass reconstruction and the target metric.** Throughout this subsection the level is admissible, so that $N = 2^a M$, $0 \le a \le 3$, and M is odd and squarefree. The shrinking-window counting application below is for $N > 1$, where the exact new target is cuspidal. At $N = 1$, the translation and Fricke equations already give full modularity, and the classical Maass window count applies directly. The letter $v > 1$ denotes a fixed weight exponent, not the spectral parameter, and

$$w(y) = \min(y/\varepsilon_0, 1)^v, \qquad d\mu = y^{-2}\, dx\, dy.$$

All constants in the statements below may depend on the fixed level, the fixed terminal height ε_0, and v.

Lemma 4.4 (Reconstruction uniform in the real spectral parameter)**.** *The weight-zero strict reconstruction certificate defines one operator R from terminal data to $L^2(w^2 d\mu)$ which is independent of the real Maass spectral parameter. It is bounded for sufficiently large fixed v, is the identity on the terminal strip, and reconstructs every Maass Euler solution from its terminal data. Consequently its Gram normalization $U = R(R^*R)^{-1/2}$ is also independent of that parameter. Every positive residual filter formed from the original weight-zero functional equations fixes the normalized terminal datum of every solution, simultaneously for all parameters.*

Proof. The weight-zero functional equations involve pullback by the same fixed rational isometries, with fixed coefficients. Their dependence on the spectral parameter lies in the Bessel expansion of the solution and not in these pullbacks. The strict height gain and the row and column Schur estimates in the reconstruction lemma therefore apply verbatim, without a spectral-parameter constant. The direct-height growth exponent is determined by the finite certificate. Taking $v > s_* + 1/2$ places each actual solution in the weighted space; its individual norm may depend on the solution, which is harmless. The normalized datum is $(R^*R)^{1/2}h$ for raw terminal datum h. Each original residual annihilates its reconstruction, so a filter $(I - S/K)^d$, with S the sum of the original residual squares, fixes this datum exactly. □

Let $\Gamma = \Gamma_0(N)$ and let $\mathcal{V}$ be the closed new/sign subspace of the unweighted quotient $L^2(\Gamma\backslash\mathbb{H})$. We work in one fixed Atkin–Lehner sign sector throughout. Write P for its orthogonal projection. Functions in $\mathcal{V}$ satisfy the fixed bad-prime equations, including $U_2F = 0$ if $4 \mid N$. We regard $\mathcal{V}$ first as an abstract closed quotient subspace, without imposing an individual Laplace eigenvalue.

Lemma 4.5 (Uniform target metric)**.** *The map*

$$J : \mathcal{V} \longrightarrow L^2((\mathbb{R}/\mathbb{Z}) \times \mathbb{R}_{>0}, d\mu), \qquad JF = wF,$$

is bounded above and below. More precisely,

$$c\|F\|^2_{L^2(\Gamma\backslash\mathbb{H})} \le \|wF\|^2_{L^2(T\backslash\mathbb{H})} \le C\|F\|^2_{L^2(\Gamma\backslash\mathbb{H})}.$$

The constants do not depend on a Laplace eigenvalue. The same statements hold in the regular quotient modules.

Proof. Put

$$s(z) = \sum_{\gamma\in T\backslash\Gamma} w(\Im\gamma z)^2.$$

Unfolding gives $\|wF\|^2 = \int_{\Gamma\backslash\mathbb{H}} s|F|^2 d\mu$. The standard elementary Eisenstein-series estimate gives $s \le C$: on a compact quotient core the series converges normally, while in each cusp the identity-type terms are bounded and the remaining terms are $O(y^{1-2v})$. Here cusp height is taken in any of the fixed scaling coordinates. This estimate also follows directly by summing $(y/|cz+d|^2)^{2v}$ over primitive bottom rows and separating $c = 0$.

For the lower bound, first observe that the admissible cusps are indexed by the divisors $d \mid N$, with one cusp for each divisor: their multiplicity is $\varphi(\gcd(d, N/d)) = 1$. An exact-divisor Atkin–Lehner involution at an odd prime toggles the corresponding denominator valuation. At 2^a it sends j to $a - j$. If $a \ge 2$, the half-translation $T_{1/2}$ normalizes Γ. Indeed conjugation sends an integral matrix $(a_0, b_0; c_0, d_0)$ to

$$\begin{pmatrix} a_0 + c_0/2 & b_0 + (d_0 - a_0)/2 - c_0/4 \\ c_0 & d_0 - c_0/2 \end{pmatrix},$$

which remains in Γ because $4 \mid c_0$ and a_0, d_0 are odd. On cusp denominators it interchanges the 2-adic valuations 0 and 1 and fixes valuations at least 2. The transformations $j \mapsto a - j$ and $0 \leftrightarrow 1$ are transitive on $\{0, \dots, a\}$ for $a = 2, 3$. Thus the fixed normalizers generated by the Atkin–Lehner maps and, when needed, $T_{1/2}$ carry every cusp to infinity. They preserve the newspace; in a fixed sign sector the relevant Atkin–Lehner actions are scalars of absolute value one. Moreover $U_2F = 0$ implies $F(z/2) + F((z+1)/2) = 0$, hence $F(z + 1/2) = -F(z)$. Choose, for each cusp, one such fixed normalizer word taking it to infinity. It maps a sufficiently high cusp tail isometrically onto an infinity cusp tail, with a fixed change in the chosen height coordinate. The square integral of F on each such tail is therefore bounded by a fixed multiple of its square integral on the infinity terminal tail, where $w = 1$. The complement of these finitely many tails is compact; there s has a positive minimum, since every term is nonnegative and at least one is positive. The compact-core bound and the cusp-tail bounds prove the lower inequality. All arguments are changes of coordinates, finite normalizer identities, and nonnegative sums, so apply equally with orbit counting measure. □

Why the full modular metric is insufficient. At a cusp not equivalent to infinity, $s(z)$ can tend to zero; one cannot assert a global positive lower bound for s on the full modular quotient. The preceding lower bound uses the new/sign equations. Nor does multiplication by $\sqrt{s}$ generally commute with P. In particular the new projection is not obtained by conjugating P by the full-modular isometry $F \mapsto wF/\sqrt{s}$.

Let $J_{\mathrm{all}}F = wF$ on the whole unweighted quotient. The actual orthogonal projection onto the weighted new/sign range is

$$Q = J_{\mathrm{all}} P\big(PsP\big|_{\mathrm{Ran}\,P}\big)^{-1} P J^*_{\mathrm{all}}. \tag{4.9}$$

The inverse in this formula is bounded by the target metric lemma. Equivalently one can use the inverse of $PsP + (I - P)$ and sandwich it by P. Its inverse and inverse square root admit exponentially convergent polynomial approximations on a fixed positive compact interval. These formulas are also valid for the regular module. They keep track of the target metric without asserting that it is diagonal in the local newspace decomposition.

4.2. **Finite branches for the modular projection after cusp truncation.** Let h_Γ be a fixed quotient cusp-height function, equal to a bounded positive function on a compact core. Its normalization is irrelevant below. Put $C_H = \mathbf{1}_{h_\Gamma \le H}$. Define the full-modular isometry and its truncated range projection by

$$J_0 F = wF/\sqrt{s}, \qquad Q_H = J_0 C_H J_0^*.$$

Here Q_H is the full modular projection with a quotient cutoff; it is not the new projection Q of (4.9).

Lemma 4.6 (Explicit finite-branch modular approximation)**.** *Let $\chi_Y = \mathbf{1}_{y \ge Y}$ on the translation cylinder. For $0 < Y \le 1 \le H$,*

$$\|Q_H - \chi_Y Q_H \chi_Y\| \le C H^{v+1} Y^{v-1}.$$

The right side may be replaced by its minimum with 2. The operator $\chi_Y Q_H \chi_Y$ is a finite sum of rational pullback branches. Its source and image heights lie in $[Y, CH]$, and the number and primitive matrix heights of those branches are bounded by a fixed power of $H + Y^{-1}$. Its coefficients are measurable functions, with bounds polynomial in the same quantities. These claims hold in both the physical and regular representations.

Proof. Choose one quotient representative for $h_\Gamma \le H$. Its translation-folded height is between c/H and CH. Consequently $s \ge cH^{-2v}$ there. The same primitive-bottom-row calculation as above, now with exponent 2, gives

$$E_2(z) := \sum_{T\backslash\Gamma} (\Im \gamma z)^2 \le CH^2 \qquad (h_\Gamma(z) \le H).$$

For Y below a fixed terminal constant, the omitted orbit mass is therefore bounded by

$$\sum_{\Im \gamma z < Y} w (\Im \gamma z)^2 \le C Y^{2v-2} E_2(z) \le C Y^{2v-2} H^2.$$

After division by s this is at most $\rho = CH^{2v+2} Y^{2v-2}$. Thus $\|(I - \chi_Y) J_0 C_H\| \le \sqrt{\rho}$; splitting the difference of the two range operators proves the displayed norm estimate.

On the retained set, a primitive determinant-one matrix $\gamma = (a, b; c, d)$ mapping z to another retained height satisfies $|cz + d|^2 = y/\Im \gamma z \le CH/Y$. Since $y \ge Y$, this bounds $|c| \le C\sqrt{H/Y}/Y$ and then $|d|$ polynomially after folding $0 \le x < 1$. For each bottom row, left translation chooses the unique image fold, up to endpoints of measure zero; its top row is also polynomially bounded because both real parts lie in $[0, 1)$ and both heights are retained. There are only polynomially many such integral matrices. The coefficient formula for the projection is

$$(J_0 J_0^* u)(z) = \frac{w(\Im z)}{s(z)} \sum_{\gamma \in T\backslash\Gamma} w(\Im \gamma z) u(\gamma z).$$

Insert the invariant cutoff C_H and the two height cutoffs in this formula. The lower bound for s and the finite matrix bounds give all assertions. Coincident folded branches can simply be combined, and boundary choices have measure zero. □

The removed quotient cusp module has regular trace $O(H^{-1})$. A hyperbolic propagation of length at most R can enlarge it only to quotient cusp height at least cHe^{-R}, whose trace is $O(e^R/H)$. This follows from the fact that the logarithm of cusp height has gradient of norm one in each sufficiently high cusp. No small global operator norm is claimed for the removed cusp module.

4.3. **An exact target–excess counting identity.** The following abstract lemma separates the simultaneous-window counting argument from its remaining analytic trace estimate. It also applies when terminal data have first been restricted to the nonzero Fourier modes and reconstruction renormalized on that closed subspace. All normalizations in a given application must use that same terminal space.

Lemma 4.7 (Counting without orthogonality across parameters)**.** *Let $U : \mathcal{G} \to \mathcal{H}$ be the reconstruction isometry, and let $\mathcal{V}$ be the cuspidal quotient new/sign space. Suppose the map $J_T : \mathcal{V} \to \mathcal{G}$ satisfies $UJ_TF = wF$ and is bounded above and below. Let M denote the orthogonal projection onto its range. Let A be a positive contraction fixing M and fixing the normalized datum of every solution under consideration. Put $E = A - M$, so $0 \le E \le I - M$.*

Let h be a real even spectral multiplier satisfying $h(r) \ge 1$ on the requested window, and put $K = wh(\sqrt{\Delta - 1/4})w^{-1}$ on the ambient cylinder. Assume it acts on every actual reconstructed solution by its scalar $h(r)$ and agrees on $w\mathcal{V}$ with the quotient wave operator. If the operators on the right below are Hilbert–Schmidt, then any finite linearly independent family of actual solution data has size at most

$$\operatorname{Tr}_{\mathcal{V}} h(\sqrt{\Delta - 1/4})^2 + C\|KUE\|_{\mathrm{HS}}^2. \tag{4.10}$$

The constant depends only on the bounds for J_T and its inverse. If $K^*K \le C_R M_t$ for a positive unit-width majorant M_t, then the second term is at most

$$CC_R \operatorname{Tr}(M_t U E^2 U^*). \tag{4.11}$$

Proof. The identity $AM = M$ and selfadjointness imply $MA = M$, so $A = M + E$ is block diagonal. Equip $M\mathcal{G}$ with the quotient norm $\|J_T^{-1}g\|$ and leave $(I - M)\mathcal{G}$ orthogonal with its old norm. This is an equivalent Hilbert norm. Explicitly its isometry to the original Hilbert norm is

$$L = (J_T J_T^*)^{-1/2} M + (I - M),$$

where the inverse is taken on $M\mathcal{G}$. Both L and L^{-1} have fixed bounds, and $LA = AL$. Define

$$B = U^*KUA, \qquad B' = LBL^{-1}.$$

Every actual datum g is an eigenvector of B with eigenvalue $h(r)$: indeed $Ag = g$, $KUg = h(r)Ug$, and $U^*U = I$. Distinct parameters need not be orthogonal for this statement.

The target space is invariant under U^*KU, since K preserves $w\mathcal{V}$. On $M\mathcal{G}$ the operator B' is unitarily equivalent to the canonical quotient multiplier on $\mathcal{V}$. On $(I - M)\mathcal{G}$ it equals LU^*KUE, because L^{-1} is the identity there. Summing squared column norms in an orthonormal basis adapted to the source decomposition gives the exact identity

$$\|B'\|_{\mathrm{HS}}^2 = \operatorname{Tr}_{\mathcal{V}} h(\sqrt{\Delta - 1/4})^2 + \|LU^*KUE\|_{\mathrm{HS}}^2.$$

There is no cross term: the two summands have orthogonal source spaces, although their image spaces need not be orthogonal.

For completeness, if a compact operator C has n linearly independent eigenvectors with eigenvalues of modulus at least one, their span is a finite-dimensional invariant space. The compression of C to that span has a Schur upper triangular matrix with those eigenvalues on the diagonal. Its Hilbert–Schmidt norm squared is at least n and at most $\|C\|_{\mathrm{HS}}^2$. Apply this to B' and use $\|U^*\| = 1$ and the fixed bound for L. Finally $\|KUE\|_{\mathrm{HS}}^2 = \operatorname{Tr}(EU^*K^*KUE)$; the positive operator inequality and cyclicity of a finite positive trace give (4.11). Truncation proves the same statement when the positive traces are initially defined by monotone limits. □

In the regular representation, if the original filter satisfies $\tau(A) - \tau(M) \le \eta$, then

$$\tau(UE^2U^*) = \tau(E^2) \le \tau(E) = \tau(A) - \tau(M) \le \eta.$$

This is the small quantity needed in (4.11); the coefficient of the exact target in (4.10) has not been enlarged.

4.4. **A unit-width trace comparison for finite rational branches.** Put

$$m_t(r) = \left(1 + \left(\frac{r^2 - t^2}{t}\right)^2\right)^{-4}, \qquad M_t = m_t(\sqrt{\Delta - 1/4}), \qquad t \geq 3.$$

The operator in this subsection is the unweighted selfadjoint Laplacian on the complete translation cylinder.

Lemma 4.8 (Unit-width finite-branch trace)**.** *Let W be a finite sum of rational pullback branches on the translation cylinder, with measurable coefficients. Suppose all source and image heights lie in $[Y, H]$, where $0 < Y \leq 1 \leq H$, all primitive integral matrices have height at most B, and the sum of the coefficient sup norms is Λ. Then, for a fixed absolute exponent C,*

$$\left|\mathrm{Tr}(M_t^{1/2} W M_t^{1/2}) - \kappa_t \tau(W)\right| \leq C\sqrt{t}\,\Lambda(1 + B + H + Y^{-1})^C, \tag{4.12}$$

where the regular trace uses hyperbolic area and

$$\kappa_t = \frac{1}{2\pi}\int_0^\infty m_t(r) r \tanh(\pi r)\, dr \ll t.$$

The identity coefficient in $\tau(W)$ is taken after folding modulo integer translations. For positive W the trace in (4.12) *is the usual positive quadratic-form trace, so the statement applies directly to a positive finite-branch approximation of UE^2U^*.*

Proof. We first record an explicit bound for the radial kernel $k_t(\rho)$ on $\mathbb{H}$:

$$|k_t(\rho)| \leq Ct(1 + t\rho)^{-1/2} e^{-5\rho/8}. \tag{4.13}$$

Here and below the harmless value of the absolute exponential constant can be decreased. To verify the bound, set $\alpha + i\beta = \sqrt{t^2 + it}$, with $\alpha, \beta > 0$. Then $\alpha \asymp t$ and $0.49 < \beta < 1/2$ for $t \geq 3$. The rational function m_t has order-four poles at $\pm\alpha \pm i\beta$. Residue evaluation of its inverse Fourier transform on $u \geq 0$ gives

$$g_t(u) = \sum_{\pm} e^{\pm i\alpha u - \beta u} \sum_{j=0}^{3} c_{j,\pm}(t) u^j, \qquad |c_{j,\pm}(t)| \leq C.$$

These coefficient bounds follow either by differentiating the pole factors, or by noting that the other nearby pole is at a distance $2\beta \asymp 1$ and the two opposite poles are at distances $\asymp t$, which cancel the numerator factor t^8. The same description for g_t' has coefficients $O(t)$.

The hyperbolic Abel inversion formula expresses $k_t(\rho)$ as a fixed constant times

$$-\int_\rho^\infty \frac{g_t'(u)}{\sqrt{\cosh u - \cosh \rho}}\, du.$$

For $\rho \geq t^{-1}$, put $u = \rho + v$ and split the integral at $v = \alpha^{-1}$. The part next to the endpoint is bounded absolutely. On the remainder integrate each exponential $e^{\pm i\alpha v}$ once by parts. For $0 < \rho \leq 1$ and $v \leq 1$ the denominator is comparable to $\sqrt{v(\rho + v)}$; its inverse and its derivative are bounded by $C\rho^{-1/2}v^{-1/2}$ and $C\rho^{-1/2}v^{-3/2}$, respectively. Thus both the endpoint part and the integrated remainder are $O(t\alpha^{-1/2}\rho^{-1/2}) = O(\sqrt{t}\rho^{-1/2})$. The portion $v \geq 1$ is absolutely integrable after the same integration by parts. For $\rho \geq 1$, use instead

$$(\cosh(\rho + v) - \cosh \rho)^{-1/2} \ll e^{-\rho/2}(e^v - 1)^{-1/2}.$$

The endpoint argument gives $O(\sqrt{t}(1 + \rho)^3 e^{-(1/2+\beta)\rho})$. Absorbing the polynomial factor, and the additional factor $\rho^{1/2}$, into the exponential proves (4.13) for $\rho \geq t^{-1}$. For smaller ρ, positivity of the

spectral density and the bound of the spherical function by one give $|k_t(\rho)| \le k_t(0) = \kappa_t \ll t$. The last bound also follows directly by substituting $s = (r^2 - t^2)/t$ in the displayed formula for κ_t.

For a primitive integral representative $g = (a, b; c, d)$, put $D = \det g > 0$ and $p_g(z) = cz^2 + (d - a)z - b$. Hyperbolic displacement satisfies

$$2\sinh(d(z, gz)/2) = \frac{|p_g(z)|}{y\sqrt{D}}.$$

If g is not scalar, this is a nonzero polynomial with integral coefficients. On $0 \le x \le 1$, $Y \le y \le H$, its sublevel area obeys

$$\operatorname{area}\{|p_g(z)| < r\} \le C(1 + H)(r + r^2).$$

For a quadratic polynomial this follows by factoring over $\mathbb{C}$: if the product of the two root distances is less than $r/|c|$, at least one distance is less than $\sqrt{r/|c|}$. The linear case is a disk of radius at most r; the nonzero constant case is empty for $r < 1$ and is bounded by the rectangle area otherwise. Since $D \le 2B^2$ and $y \le H$, this implies a bound $\mu\{d(z, gz) < r\} \le C(1 + B + H + Y^{-1})^C r$ for $0 < r \le 1$. Layer-cake integration, using $\rho^{-1/2}$ for small displacement, now gives

$$\int_{\text{branch domain}} |k_t(d(z, gz))|\, d\mu(z) \le C\sqrt{t}(1 + B + H + Y^{-1})^C.$$

The still smaller region $\rho < t^{-1}$ contributes $O(1)$ times the same polynomial, by the sublevel estimate and the bound Ct.

The cylinder kernel is the periodization over T_n. Choose the image fold so that both source and image real parts lie in $[0, 1)$. For $|n| \le C(1 + H)$ there are polynomially many terms; the primitive matrices $T_n g$ have polynomially bounded heights and unchanged determinants, so the preceding argument applies. For the remaining terms, both endpoint heights are at most H and the horizontal displacement is at least $|n| - 1$. Hence $d(z, T_n gz) \ge 2\log(|n|/(CH))$ after increasing the cutoff. Equation (4.13) bounds their sum by

$$C\sqrt{t} \sum_{|n| > C(1+H)} (H/|n|)^{5/4},$$

times the polynomial bound for the domain area. This is again a polynomial geometric cost times $\sqrt{t}$. When a periodized branch is the identity, its zero-displacement term is exactly κ_t times its coefficient integral; the other translates satisfy the same error estimate. These are precisely the identity coefficients selected by the regular trace.

Finally multiply by the coefficient sup norm and sum the branches. Measurable coefficients cause no differentiability loss because only their absolute values enter the kernel integrals. The trace formula can be justified first for finite cutoffs of the positive spectral operator and simple coefficient functions, then by dominated convergence using the estimates just proved. Equivalently, $\chi M_t^{1/2}$ is Hilbert–Schmidt for every compact height cutoff χ: its squared norm is the integral of the cylinder diagonal, bounded by $\kappa_t \mu(\operatorname{supp}\chi)$ plus the finite periodization error just estimated; the two compact endpoint cutoffs make the sandwiched branch trace-class. This proves (4.12). □

Lemma 4.9 (Insertion of the zero-mode projection)**.** *The finite-branch trace estimate remains valid for finite words with insertions of*

$$\Pi = I - \Pi_0, \qquad (\Pi_0 f)(x + iy) = \int_0^1 f(x + u + iy)\, du.$$

In calculating the main coefficient, replace Π *by* I*. The error cost is polynomial in the compact endpoint geometry and in a bound for the norms and inverse norms of all determinant-one partial matrix products in each word, multiplied by the coefficient sum after expansion. In particular, a word involving a* Π_0 *has no identity main term.*

Proof. Expand every Π as $I - \Pi_0$. For a term containing an averaging factor, fix all averaging parameters except one. The resulting matrix has the form $gT_u h$, $0 \le u \le 1$, with real determinant-one g, h whose norms and inverse norms have the stated bound. Its displacement polynomial is $p_0(z) + up_1(z)$. The matrix giving p_1 is $gE_{12}h$, a nonzero rank-one real matrix. Writing it as a column a times a row b, one has

$$p_1(z) = (b_1 z + b_2)(a_2 z - a_1).$$

For a real pair (c, d) and $Y \le \Im z \le H$, $0 \le \Re z \le 1$, $|cz + d|$ is bounded below by a fixed negative power of $1 + H + Y^{-1}$ times $\sqrt{c^2 + d^2}$. The column and row norms are bounded below by a fixed negative power of the matrix bound. Thus $|p_1(z)|$ has a uniform polynomial lower bound on the retained rectangle. For each z, the set of parameters for which $|p_0(z) + up_1(z)| < r$ is an interval of length at most a polynomial geometric constant times r. This proves the same linear displacement sublevel bound in the joint (z, u) variables as in the preceding lemma. Its kernel integration and periodization argument therefore give the $O(\sqrt{t})$ estimate. Identity matrices can occur for at most one value of u and have measure zero. Additional averaging parameters are integrated afterwards. Measurable coefficient functions and branch indicators are bounded absolutely before this argument, so their dependence on the parameters causes no change. □

The finite-branch estimate also permits orientation-reversing rational isometries, and therefore the fixed parity projection. Write such an isometry as $(a\overline{z} + b)/(c\overline{z} + d)$ with an integral matrix of negative determinant. Its displacement numerator is

$$p(z) = c|z|^2 + (d - a)x - b + i(d + a)y.$$

Its sublevel area is still linear in r, with polynomial dependence on H, Y^{-1}. If $d + a \ne 0$, the sublevel set is empty for $r < Y$, and the trivial rectangle bound suffices for $r \ge Y$. If $d = -a$ and $c \ne 0$, its real part is monotone in y, with derivative of absolute value at least $2Y$; each vertical sublevel interval has length at most r/Y. If also $c = 0$, negative determinant forces $a \ne 0$, and the x derivative has absolute value at least 2. The kernel and periodization proof therefore gives the same $O(\sqrt{t})$ error, with no identity main term. In the zero-mode insertion argument the rank-one factorization becomes a product of a real linear factor in z and one in $\overline{z}$; the same lower bound applies. All subsequent estimates consequently include the fixed parity projection.

4.5. **Trace approximation for the entire excess.** For the Maass argument choose a smooth weight comparable to w, equal to $(y/\varepsilon_0)^v$ for $y \le \varepsilon_0/2$ and to 1 for $y \ge \varepsilon_0$, with bounded covariant derivatives of its logarithm of every fixed order. All preceding reconstruction and geometric estimates persist with changed fixed constants. Choose v at least two larger than an exponent for which the reconstruction Schur bound holds.

Use the terminal Hilbert space $\mathcal{G}_! = \Pi\mathcal{G}$ and the isometry

$$U = R\Pi\big(\Pi R^* R\Pi\big|_{\mathcal{G}_!}\big)^{-1/2}.$$

The identity terminal component of R ensures that the compressed Gram operator is at least I. An actual solution has zero constant Fourier term, so this normalization preserves all exact solution identities. In the regular representation Π is interpreted as I: indeed the projections averaging the finite groups of dyadic translations converge strongly to zero in the regular representation and to Π_0 on the physical cylinder. The preceding lemma verifies this interpretation directly at the required spectral-trace level.

Including parity in the original filter. For a fixed parity σ, include the known equation

$$F(z) - (-1)^\sigma F(-\overline{z}) = 0$$

in the initial observation family J_0. Observe it on the same compact rectangles used for the other initial residuals. Reflection preserves height, so this adds only fixed or polynomial geometry costs.

On the quotient, reflection is a unitary involution commuting with the local-new and Atkin–Lehner projections. The positive penalty $I-(-1)^{\sigma}\mathcal{R}$ has spectral gap 2 off the desired parity sector (the unscaled residual square has gap 4). Thus the target observability and collective filter arguments apply with the exact target $M=M_{\sigma}$, and give

$$\tau(A_X)-\tau(M_{\sigma})\ll(\log X)^{-b}.$$

The reflection matrix is integral and invertible modulo every prime. Its addition to the fixed initial observations creates no further fresh-prime exclusions and changes neither the rank contraction nor the collective triangular determinant argument. Its orientation-reversing trace contributions are covered above. This parity observation is essential: merely replacing the target by its parity subspace without adding the corresponding known equation would not imply the displayed regular excess estimate.

Proposition 4.10 (Quantitative unit-width excess trace)**.** *Let A_X be the single positive filter furnished by the collective regular argument, rebuilt using the zero-mode-free isometry above. Write $\ell=\log X$. Suppose its established bounds are*

$$\tau(A_X)-\tau(M)\ll\ell^{-b},\qquad \log d_X\ll\ell^{1+b},\qquad 0<b<1,$$

where d_X is its degree and its original residual norms and branch complexities are polynomial in X. Then, with $E_X=A_X-M$,

$$\operatorname{Tr}(M_tUE_X^2U^*)\ll t\ell^{-b}+\sqrt{t}\exp\bigl(\exp(C\ell^2)\bigr),\qquad t\geq 3. \tag{4.14}$$

Here one uses the physical and regular local-new projection gap from the separate local-gap lemma, and the nonzero-channel cusp bound

$$\operatorname{Tr}(M_t\mathbf{1}_{y>H}\Pi)\ll t/H. \tag{4.15}$$

The latter is the Kontorovich–Lebedev cusp estimate and applies for every $H\geq\varepsilon_0$.

Proof. Put $W=UE_X^2U^*$. It is positive, $W\leq UU^*$, and $\tau(W)\ll\ell^{-b}$. The exact target projection Q satisfies $UMU^*=Q$. Since $A_XM=M$, there is the useful exact identity

$$W=UA_X^2U^*-Q. \tag{4.16}$$

Thus it suffices to approximate the target projection linearly; no positivity assertion for its polynomial approximants is needed.

Choose $Y=e^{-D\ell^2}$ and $H=e^{D\ell^2}$, for a sufficiently large fixed D, and let $\chi=\mathbf{1}_{Y\leq y\leq H}$. Comparison with reconstruction at exponent $v-2$ gives the ambient operator inequality

$$UU^*\leq C\min(y/\varepsilon_0,1)^4.$$

This follows by factoring the two powers of the weight from R and using the compressed Gram inverse bound. The periodized kernel bound above then gives

$$\operatorname{Tr}(M_t\mathbf{1}_{y<Y}W\mathbf{1}_{y<Y})\ll tY^3.$$

For clarity, the identity kernel term integrates $Ct\int_0^Y y^4dy/y^2$; the nonidentity translations contribute at most $C\sqrt{t}\int_0^Y y^4y^{5/4}dy/y^2$. On the high terminal strip the range of U is in the nonzero Fourier modes. Hence its high compression is bounded by $C\Pi$, and (4.15) bounds the corresponding high trace by Ct/H. The low and high pieces together are $O(t(Y^3+H^{-1}))$, by the Hilbert–Schmidt triangle inequality.

We describe the compact approximation, in both representations. Truncate reconstruction deeper than Y, at $Y_0=e^{-D_0\ell^2}$ with a larger fixed D_0, and approximate its compressed Gram inverse square root by a polynomial. The discarded reconstruction norm is a fixed positive power of Y_0; the Gram approximation degree is $O(\ell^2)$. The single-filter telescoping bound is $O(d_X)$ times the reconstruction

error, so D_0 can be chosen to make the compact norm error as small as $\zeta = e^{-D_1\ell^2}$ for any prescribed fixed D_1. The same coefficient and height bookkeeping as in the single-filter holomorphic transfer gives

$$\log(2 + \Lambda + B + \text{partial-product bounds}) \le Cd_X(1+\ell)^C \le \exp(C\ell^2).$$

Insertions of Π in the compressed Gram polynomials are covered by the preceding zero-mode lemma; expanding them only changes the constant in this bound.

For the target term in (4.16), use (4.9). The local-new gap gives polynomials P_m in the fixed local averaging operators, converging to P in both physical and regular operator norm at rate Ce^{-cm}, with branch complexity e^{Cm}. Include the fixed sign projections in these polynomials. The operator $PsP + (I - P)$ has spectrum in a fixed positive compact interval by the target metric lemma. Its inverse is uniformly approximable by polynomials of degree $O(\log \zeta^{-1})$. Replacing P by P_m with $m = O(\log \zeta^{-1})$, and increasing that constant to absorb polynomial sensitivity, gives an approximation to Q with error $O(\zeta)$. The bounded function s is an allowed measurable coefficient and need not itself be expanded as an Eisenstein series.

The factors $J_{\text{all}}^*\chi$ and χJ_{all} have finite branch descriptions. Indeed the quotient cusp height of a cylinder point with $Y \le y \le H$ is bounded by a fixed power of $H + Y^{-1}$; for any one of the fixed cusp scalings this follows from $\Im(gz) = y/|cz + d|^2$, treating $c = 0$ separately. The same primitive-matrix bounds used for Q_H then apply to the endpoint-cut orbit sums. Each word of length $O(m)$ in the local operators changes cusp height by at most e^{Cm}. The inverse-polynomial composition therefore has logarithmic branch complexity $O(\ell^C)$, and all intermediate coordinates lie in fixed powers of $e^{C\ell^C}$. This cost is smaller than the single-filter cost already displayed.

We have thus obtained a compact finite-word operator W_f such that

$$\|\chi W \chi - W_f\| \le C\zeta$$

in both representations, with endpoints in $[Y, H]$ and total finite-branch trace cost at most $\exp(\exp(C\ell^2))$. The trace error from the norm approximation is at most $C\zeta \operatorname{Tr}(M_t\chi)$; the regular trace error is at most $C\zeta\tau(\chi) \ll \zeta/Y$. The unit-width finite-branch and zero-mode lemmas consequently give

$$\operatorname{Tr}(M_t\chi W\chi) \le Ct\big(\ell^{-b} + \zeta/Y\big) + \sqrt{t}\exp(\exp(C\ell^2)),$$

since $\tau(\chi W\chi) \le \tau(W)$.

Finally write $M_t^{1/2}W^{1/2}$ as the sum with the inserted cutoffs χ and $I - \chi$. The squared Hilbert–Schmidt norm is bounded by twice the sum of their squared norms. The preceding tail estimates prove (4.14), after choosing $D_1 > D + 1$. The fixed factor two is harmless for this excess estimate; it has never been applied to the exact target trace in the counting lemma. □

Parameters for the shrinking window. Combining (4.14) with the weighted wave majorant gives the excess contribution

$$\ll e^{CR}\left\{t(\log X)^{-b} + \sqrt{t}\exp(\exp(C(\log X)^2))\right\}.$$

Choose $\log X = a\sqrt{\log\log t}$ with a sufficiently small fixed $a > 0$. Let the width satisfy $0 < \varepsilon(t) \le 1$ and the hypothesis $\varepsilon(t)\log\log\log t \to \infty$. Put

$$R = \sqrt{\frac{\log\log\log t}{\varepsilon(t)}}.$$

Then $R\varepsilon \to \infty$, $R = o(\log\log\log t)$, and the entire displayed excess is $o(t\varepsilon)$. The exact target contribution is the classical newform trace of the chosen Paley–Wiener majorant squared. Its asymptotic is the genuine newform window coefficient, provided that majorant has squared mass $\varepsilon + o(\varepsilon)$, as in the separate spectral-majorant lemma. For the total count one sums the finitely many fixed sign sectors only after applying the target metric and counting arguments in each sector.

4.6. **A physical as well as regular gap for the local new projection.** This subsection supplies an input that cannot be inferred from the regular Bruhat–Tits tree calculation alone. We use the uniform unramified Satake bound

$$p^{-\theta} \le |\alpha_{j,p}| \le p^{\theta}, \qquad \theta = 7/64, \tag{4.17}$$

for cuspidal representations of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$. This is the Kim–Sarnak bound [4, Appendix 2]; see also [7, equation (23)]. Any established uniform exponent $\theta < 1/2$ suffices below. The exponent is used only at the finitely many fixed ramified primes.

Let N be admissible as in the main manuscript. On the full quotient $L^2(\Gamma_0(N)\backslash\mathbb{H})$, let E_p^0, E_p^1 be the orthogonal averages onto the two adjacent old spaces, with the same normalizations as in the local new-trace calculation. Put

$$S_p = E_p^0 + E_p^1, \qquad P_p = 1_{\{0\}}(S_p) = I - (E_p^0 \vee E_p^1).$$

The identical definitions apply in the regular module.

Lemma 4.11 (Uniform physical local-new gap)**.** *In both the physical quotient representation and the regular module,*

$$S_p \ge g_{p,a}(I - P_p), \qquad \|S_p\| \le 2, \quad a = v_p(N),$$

where one may take

$$g_{p,1} = 1 - \frac{\sqrt{p}\,(p^{\theta} + p^{-\theta})}{p+1} > 0,$$
$$g_{2,2} = 1/2, \qquad g_{2,3} = 1.$$

The statement includes the continuous spectrum and the constant representation; neither is silently discarded.

Proof. Suppose first $a = 1$. In the local automorphic decomposition the $K_0(p)$-fixed vectors consist of the unramified old part and the conductor-one new part. On the latter both old averages vanish. On an irreducible unramified generic constituent the two old ranges are lines spanned by normalized spherical vectors v_0, v_1 based at the endpoints of an edge. If T_p denotes the normalized spherical Hecke operator and $T_p v_0 = \lambda_p v_0$, then

$$\langle v_1, v_0 \rangle = \frac{\sqrt{p}}{p+1}\lambda_p. \tag{4.18}$$

Indeed, all $p+1$ neighboring translates of v_0 have the same matrix coefficient against v_0, and their sum divided by $\sqrt{p}$ is $T_p v_0$. Thus the sum of the two rank-one projections has eigenvalues $1 \pm |\sqrt{p}\lambda_p/(p+1)|$ on their span.

For a cuspidal constituent with trivial central character, $\lambda_p = \alpha_p + \alpha_p^{-1}$ is real. The unitary spherical classification and (4.17) give $|\lambda_p| \le p^{\theta} + p^{-\theta}$; this also covers the tempered case, where $|\lambda_p| \le 2$. For continuous Eisenstein constituents the inducing characters are unitary, so the same estimate holds with $\theta = 0$ whenever an unramified old line occurs. A one-dimensional residual constituent requires separate treatment: the two lines coincide, so $E_p^0 = E_p^1 = I$ and $S_p = 2I$. There is no small positive eigenvalue arising by substituting the trivial Satake parameter into the two-dimensional formula. In the present trivial-character quotient this residual constituent is the constant representation. Integration of these uniform constituent estimates proves the physical inequality. In the regular module the old-line angle is at most $2\sqrt{p}/(p+1)$, so the same, slightly weaker, $g_{p,1}$ applies there as well.

Now let $p = 2$ and $a = 2$. Conjugate the local compact group generated by the two old groups so that it is $K = \mathrm{GL}_2(\mathbb{Z}_2)$ (or the determinant-one version). The intersection subgroup fixing the two outer vertices reduces modulo 2 to the diagonal subgroup, which is trivial over $\mathbb{F}_2$. Thus the

relevant action factors through $\mathrm{GL}_2(\mathbb{F}_2) \cong S_3$ on its six-element regular coset space. The two old subgroups reduce to two distinct order-two subgroups generated by transpositions s, t. Consequently

$$S_2 = \frac{I+s}{2} + \frac{I+t}{2} = I + \frac{s+t}{2}.$$

On the trivial representation its eigenvalue is 2, on the sign representation it is 0, and on the two-dimensional standard representation its eigenvalues are $1/2$ and $3/2$. Every unitary representation of this finite group has the same spectral inclusion, proving the gap $1/2$ in either ambient model.

For $p = 2$ and $a = 3$, the group generated by the two old groups is, after conjugation, the Iwahori group $K_0(2)$. Their intersection is

$$H = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in K_0(2) : 2 \mid b,\ 4 \mid c \right\}.$$

The map

$$K_0(2) \longrightarrow \mathbb{F}_2^2, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto (b \bmod 2, c/2 \bmod 2)$$

is a surjective homomorphism: a, d are odd, so matrix multiplication adds the two displayed coordinates. Its kernel is H, and the two old subgroups are the inverse images of the two coordinate lines. Their averages therefore commute and S_2 has spectrum contained in $\{0, 1, 2\}$. This proves the last gap and finishes the proof. □

Corollary 4.12 (Simultaneous finite-branch approximation)**.** *Let $P = \prod_{p|N} P_p$, and optionally multiply P by any fixed commuting Atkin–Lehner and reflection sign projectors. There exist positive finite-branch contractions $P^{(\ell)}$ such that, simultaneously in the regular and physical representations,*

$$P^{(\ell)} P = P, \qquad \|P^{(\ell)} - P\| \le \omega(N) e^{-c_N \ell}.$$

The branch count, total absolute coefficient, and primitive rational matrix heights are at most $\exp(C_N \ell)$.

Proof. Use

$$P^{(\ell)} = \prod_{p|N} (I - S_p/2)^{\ell},$$

and insert the same optional sign projectors. The local factors at distinct primes commute. Atkin–Lehner reversal interchanges the two old averages and hence commutes with their sum; the other sign actions also commute with all the factors. On $I - P_p$, the corresponding factor is at most $(1 - g_{p,a}/2)^{\ell}$ in norm, while it is exactly the identity on P_p. A telescoping product estimate gives the stated error, with $c_N = \min_{p|N} - \log(1 - g_{p,a}/2) > 0$. Each old average and sign action has finitely many fixed rational branches. Expanding a product of $O_N(\ell)$ such operators gives all three complexity bounds. For $N = 1$ the empty product is the identity and no arithmetic gap is required. □

Scope of this addition. This is a gap on the automorphic quotient and on the regular module. It is not asserted on the space of arbitrary nonmodular functions on the translation cylinder. Its role there is only through the modular embedding and the correctly compressed target metric $PsP|_{\mathrm{Ran}\, P}$. Approximation of that metric by polynomials must use its separate positive lower bound, rather than presume that multiplication by s commutes with P.

5. Weighted wave operators

5.1. **A weighted wave operator and a unit-width spectral majorant.** Let $X = \langle T_1 \rangle \backslash \mathbb{H}$, let $\Delta \ge 0$ be its self-adjoint hyperbolic Laplacian on $L^2(X, d\mu)$, and let $w > 0$ be smooth. Assume the covariant derivatives of $\log w$ of orders 1 through 16 are bounded. A smooth replacement of

$\min(y/\varepsilon_0, 1)^\alpha$, equal to it outside a fixed transition region, has this property. The replacement is uniformly equivalent to the original weight.

Lemma 5.1 (Finite propagation and weight bands)**.** *Suppose T is bounded on $L^2(X)$ and has propagation at most R. If $\|\nabla \log w\|_\infty \le L$, then*

$$\|wTw^{-1}\| \le C(1+LR)e^{LR+3}\|T\|.$$

The conjugated operator, initially defined on vectors supported in finitely many weight bands, extends boundedly. The same estimate holds with w and w^{-1} interchanged.

Proof. Let P_n multiply by $1_{\{n\le \log w<n+1\}}$. Finite propagation and the Lipschitz bound give $P_nTP_m = 0$ if $|n-m| > LR+2$. Every remaining block of the conjugated operator has norm at most $e^{LR+3}\|T\|$: indeed on the two bands the norms of multiplication by w and w^{-1} are bounded by e^{n+1} and e^{-m}, respectively. Set $J = \lceil LR+2\rceil$. For a vector $f = \sum_m P_m f$ supported in finitely many bands, Cauchy–Schwarz gives

$$\begin{aligned}\|wTw^{-1}f\|^2 &\le (2J+1)e^{2LR+6}\|T\|^2 \sum_n \sum_{|m-n|\le J} \|P_m f\|^2 \\ &= (2J+1)^2 e^{2LR+6}\|T\|^2\|f\|^2.\end{aligned}$$

Taking square roots gives the assertion. The finite-band subspaces have dense union because $0 < w < \infty$ everywhere, so the extension is unique. Replacing $\log w$ by $-\log w$ proves the last claim. □

For $t \ge 2$ put

$$\lambda_t = 1/4 + t^2, \qquad L_t = (\Delta - \lambda_t)/t, \qquad M_t = (1+L_t^2)^{-4}.$$

We use the angular Fourier convention $\widehat{h}(s) = \int_{\mathbb{R}} h(r)e^{-irs}\,dr$; thus support in $[-R,R]$ gives wave propagation at most R. Let $h = h_{t,R}$ be an even entire Paley–Wiener multiplier with propagation radius at most $R \ge 1$. In particular $H = h(\sqrt{\Delta - 1/4})$ has that propagation by the wave equation. We require the following precise scalar localization bounds:

$$\sup_{\lambda\ge 0}(1+\lambda)^{l/2}\left|\left(\frac{\lambda-\lambda_t}{t}\right)^a h\big(\sqrt{\lambda-1/4}\big)\right| \le Ce^{CR}t^l \qquad (0\le a,l\le 4). \tag{5.1}$$

Constants may depend on a fixed finite family of Schwartz seminorms, but are independent of t and R. The branch of the square root is immaterial because h is even. A uniformly bounded smooth window multiplier supported spectrally near $\pm t$, with unit-width Schwartz tails and Fourier support in $[-R,R]$, satisfies these bounds. On the short imaginary segment, the Paley–Wiener factor $e^{O(R)}$ is permitted. The estimate needed there is explicitly $h(iu) \ll_A e^{CR}t^{-A}$ for $|u| \le 1/2$, with A fixed large enough; it holds for the usual translated bandlimited windows.

Proposition 5.2 (Weighted unit-width majorant)**.** *Under* (5.1)*, the operator $K = wHw^{-1}$ satisfies*

$$\|K(1+L_t^2)^2\| \le C_w e^{C_w R}, \qquad K^*K \le C_w e^{C_w R}M_t.$$

The first expression, initially defined on smooth spectral-core vectors, has a bounded extension. Consequently, for any bounded operator $F : \mathcal{E} \to L^2(X)$ for which the right side below is finite,

$$\|KF\|_{\mathrm{HS}}^2 \le C_w e^{C_w R}\operatorname{tr}(F^*M_tF) = C_w e^{C_w R}\operatorname{tr}(M_tFF^*). \tag{5.2}$$

These are ordinary Hilbert-space traces; no regular-to-ordinary trace comparison is part of this analytic lemma.

Proof. We first fix the operator domains independently of the estimates below. The cylinder is complete. On the bundle of exterior forms the Hodge–Dirac operator $\mathcal{D} = d + d^*$ is formally symmetric, has smooth coefficients, and has propagation speed one, since the norm of its principal symbol at a covector ξ is $|\xi|$. The complete-manifold powers theorem of Chernoff therefore makes every power of $\mathcal{D}$ essentially self-adjoint on compactly supported smooth forms [2]. The even powers preserve form degree. Projecting a graph-core approximation onto degree zero proves that $C_c^\infty(X)$ is a core for every integer power of the scalar Laplacian, in particular for Δ^4 and $(1 + L_t^2)^2$. This cited domain result is used before, rather than deduced from, the energy estimates.

Two standard differential facts needed below have uniform constants on this cylinder, even though its injectivity radius tends to zero. First, if D is a differential operator of order at most l with bounded covariant coefficients, then

$$\|Df\|_2 \le C_D \|(1 + \Delta)^{l/2} f\|_2. \tag{5.3}$$

For the integral orders needed here, start with $\|\nabla f\|_2^2 = \langle \Delta f, f\rangle$ for $f \in C_c^\infty(X)$. In the integral of $|\nabla^l f|^2$, move the derivatives from one factor to the other and reorder them in contracted pairs. Each interchange produces a curvature commutator; the curvature is -1 and its covariant derivatives vanish, so each such term has at least two fewer derivatives and constant bounded tensor coefficients. Repeating this reduction leaves a finite linear combination of $\langle \Delta^j f, f\rangle$, $0 \le j \le l$. The number and coefficients of these terms depend only on l. The spectral theorem bounds their absolute sum by $C_l\langle (1 + \Delta)^l f, f\rangle$. Every D under consideration is a finite contraction of these derivatives with bounded coefficients, proving (5.3). There are no boundary integrals for compactly supported data. Closure extends the estimate to the corresponding spectral Sobolev domain. This global argument uses the curvature bound, not a positive lower injectivity radius.

Second, put $a = w^{-1}$ and $\mathrm{ad}_\Delta(A) = \Delta A - A\Delta$. For $0 \le l \le 4$ one has

$$\mathrm{ad}_\Delta^l(a) = aD_l,$$

where D_l is a differential operator of order at most l with bounded covariant coefficients. More explicitly, $D_0 = I$ and the product rule gives the recurrence

$$D_{l+1} = a^{-1}[\Delta, a]D_l + [\Delta, D_l], \qquad a^{-1}[\Delta, a]f = a^{-1}(\Delta a)f - 2a^{-1}\langle \nabla a, \nabla f\rangle.$$

The first term has order at most $l + 1$. In the second, the order-$l + 2$ symbols cancel since the principal symbol of Δ is scalar; its order is again at most $l+1$. The identities $a^{-1}\nabla a = -\nabla \log w$ and the product rule express all coefficients as polynomials in derivatives of $\log w$. After l commutators, derivatives of order at most $2l$ suffice. Taking the formal adjoint differentiates coefficients at most l more times; conjugating by w uses derivatives up to order l. Thus the sixteen-derivative hypothesis more than covers the bounded coefficients of $\widetilde{D}_l = wD_l^* w^{-1}$ for $l \le 4$. The adjoint is with respect to hyperbolic volume.

The exact normal-ordering identity is

$$a(\Delta - \lambda_t)^j = \sum_{l=0}^{j} (-1)^l \binom{j}{l} (\Delta - \lambda_t)^{j-l} \mathrm{ad}_\Delta^l(a). \tag{5.4}$$

For example $aB = Ba - [B, a]$, and induction gives the displayed binomial formula. With $H_q = L_t^q H$ it yields, for $0 \le j \le 4$,

$$KL_t^j = \sum_{l=0}^{j} (-1)^l \binom{j}{l} (wH_{j-l}w^{-1}) D_l/t^l.$$

These identities first hold on $C_c^\infty(X)$, where multiplication by $w^{\pm 1}$ and the differential operators is defined. Wave propagation and smoothness keep all intermediate expressions within the appropriate

local domains. Polynomial factors in Δ differentiate the output without enlarging its support. Every summand has norm at most $C_w e^{C_w R}$. To see this, take its adjoint and write it exactly as

$$t^{-l} D_l^* w^{-1} H_{j-l}^* w = t^{-l} w^{-1} (\widetilde{D}_l H_{j-l}^*) w.$$

The operator in parentheses has propagation at most R: neither the differential operator $\widetilde{D}_l$ nor the polynomial factor L_t^{j-l} increases propagation. Its ordinary L^2 norm is at most $Ce^{CR} t^l$ by (5.3), the spectral theorem, and (5.1). Lemma 5.1, now with weight w^{-1}, bounds the conjugated operator by $C_w e^{C_w R} t^l$. The factor t^{-l} cancels this last power. There are at most five terms, with fixed binomial coefficients. Hence the resulting bounds extend each KL_t^j from its dense smooth core to all of $L^2(X)$, with constants uniform in t.

We have proved $\|KL_t^j\| \le C_w e^{C_w R}$ for $0 \le j \le 4$. Taking the combination $K(I + 2L_t^2 + L_t^4)$ proves the first claim. If B denotes its bounded extension, then $K = B(1 + L_t^2)^{-2} = BM_t^{1/2}$ on a dense core and hence everywhere. Thus $K^*K = M_t^{1/2} B^* B M_t^{1/2} \le \|B\|^2 M_t$. The Hilbert–Schmidt bound follows by inserting F and taking positive traces. Concretely, for an orthonormal basis (e_j) of $\mathcal{E}$,

$$\sum_j \|KFe_j\|^2 \le C_w e^{C_w R} \sum_j \|M_t^{1/2} F e_j\|^2.$$

The right side is finite by hypothesis, so this calculation both proves the Hilbert–Schmidt property and its bound. The two trace expressions are the squared Hilbert–Schmidt norms of an operator and its adjoint; equivalently they follow by finite-rank truncation and monotone convergence. No cyclic rearrangement of a divergent trace is involved. □

How the estimate enters the packet argument. If U is the common reconstruction isometry, Q is its modular cuspidal-new target projection, and A is a positive contraction with $AQ = Q$, write $E = A - Q$. Then $EQ = QE = 0$ and $\tau(E^2) \le \tau(E)$. Applying (5.2) with $F = UE$ bounds the excess column of the wave-filter operator by

$$Ce^{CR} \operatorname{tr}(M_t U E^2 U^*).$$

A uniformly equivalent change of Hilbert norm on the target only multiplies this bound by a fixed constant. The arithmetic input, proved in (4.14), is a unit-width trace bound of the form

$$\operatorname{tr}(M_t U E^2 U^*) \ll t\tau(E^2) + \text{controlled approximation/trace error}.$$

This is the required location of the factor e^{CR}: it multiplies the small excess, and does not multiply the canonical target's leading Weyl coefficient.

For example, if $\tau(E) \ll (\log\log t)^{-b}$, take a window multiplier with $R\varepsilon \to \infty$ and $R = o(\log\log\log t)$. Then

$$te^{CR}\tau(E) = o(t\varepsilon)$$

under $\varepsilon \log\log\log t \to \infty$. A concrete compatible choice is $R = \sqrt{(\log\log\log t)/\varepsilon}$. The parameter choice in Section 6.1 makes the finite-branch errors small enough after multiplication by e^{CR}.

6. Window localization

6.1. **The shrinking-window test function and the final parameter choice.** Put $L = \log\log\log t$. Suppose $0 < \varepsilon = \varepsilon(t) \le 1$ and $\varepsilon L \to \infty$. Choose

$$R = \sqrt{L/\varepsilon}, \qquad b = \sqrt{\varepsilon/R}.$$

The four limits needed below follow from the exact identities

$$\frac{R}{L} = (\varepsilon L)^{-1/2}, \qquad R\varepsilon = (\varepsilon L)^{1/2}, \qquad \frac{b}{\varepsilon} = (R\varepsilon)^{-1/2}, \qquad Rb = (R\varepsilon)^{1/2}.$$

Thus $R = o(L)$, $R\varepsilon \to \infty$, $b = o(\varepsilon)$ and $Rb \to \infty$. Also $R \to \infty$, since $\varepsilon \le 1$. All assertions below are for sufficiently large t; in particular $b < \varepsilon/4$, $t > 3$, and $\varepsilon \ge L^{-1}$. No continuity or monotonicity of ε is used.

Construction and normalization of the majorant. Fix a nonnegative even Schwartz function K of integral one whose Fourier transform is supported in $[-1,1]$, and put $K_R(u) = RK(Ru)$. Let

$$I_b = [t-b, t+\varepsilon+b] \cup [-t-\varepsilon-b, -t+b],$$

$$\alpha_t = \int_{|u|>Rb} K(u)\,du, \qquad g = \mathbf{1}_{I_b} * K_R, \qquad h = (1-\alpha_t)^{-1} g.$$

For each fixed $A > 0$, Schwartz decay gives $\alpha_t \ll_A (Rb)^{-A}$, so $\alpha_t = o(1)$. For $r \in [t, t+\varepsilon]$ and $|u| \le b$ one has $r - u \in [t-b, t+\varepsilon+b]$. Consequently $g(r) \ge \int_{-b}^{b} K_R(u)\,du = 1 - \alpha_t$ and $h(r) \ge 1$. Positivity and the unit integral of K_R give $0 \le g \le 1$ and $0 \le h \le (1-\alpha_t)^{-1}$ on the real axis. Both functions are real and even. Their Fourier transforms are supported in $[-R, R]$, so the wave operator with multiplier h has propagation at most R.

Here is the weighted mass calculation, including the error caused by smoothing. Put $f = \mathbf{1}_{I_b}$. For each translation u, the symmetric difference of I_b and $I_b + u$ has length at most $4|u|$ and is contained in $|r| \le t + 3 + |u|$. Tonelli therefore gives

$$\begin{aligned}\int_{\mathbb{R}} |r|\,|g(r) - f(r)|\,dr &\le 4 \int_{\mathbb{R}} |u|(t+3+|u|) K_R(u)\,du \\ &\ll t/R + R^{-2}.\end{aligned}$$

Since $0 \le f, g \le 1$ and $f^2 = f$, we also have $|g^2 - f| \le 2|g - f|$. On the positive half-line, $\int r f(r)\,dr = (\varepsilon + 2b)(t + \varepsilon/2)$. Dividing by $(1-\alpha_t)^2$ proves

$$\int_0^\infty h(r)^2 r\,dr = t\varepsilon\left(1 + O\big(\alpha_t + b/\varepsilon + (R\varepsilon)^{-1}\big)\right) + O(1) = t\varepsilon + o(t\varepsilon).$$

The final $O(1)$ is harmless because $t\varepsilon \ge t/L \to \infty$.

The variation estimate is equally explicit: convolution with a probability density does not increase total variation, so $\mathrm{Var}(g) \le \mathrm{Var}(f) = 4$ and

$$\mathrm{Var}(h^2) \le 2(1-\alpha_t)^{-2}\,\mathrm{Var}(g) \le C.$$

For r outside I_b, integration of the Schwartz bound for K_R gives $|h(r)| \ll_A (1 + R\,\mathrm{dist}(r, I_b))^{-A}$. On $|\operatorname{Im} r| \le 1/2$, its Fourier integral gives the corresponding estimate with a factor e^{CR}; in particular $h(iu) \ll_A e^{CR} t^{-A}$ uniformly for $|u| \le 1/2$. These estimates verify (5.1): near $\pm t$, $|(r^2 - t^2)/t| \ll 1 + ||r| - t|$, while away from those intervals every fixed polynomial in r is absorbed by a sufficiently high Schwartz seminorm. The same conclusion on the imaginary segment follows from the last displayed bound.

From local counts to the smoothed genuine trace. For the genuine new spectrum, the local Weyl estimates proved above and Stieltjes integration by parts therefore give

$$\sum_{r_j \in \mathrm{new}(N)} h(r_j)^2 = \frac{\nu(N)}{6} t\varepsilon + o_N(t\varepsilon). \tag{6.1}$$

For a fixed parity, the coefficient is $\nu(N)/12$. For the total spectrum one can use the quantified global remainder $O_N(t/\log t)$. For parity, only a local remainder has been proved, so proceed as follows. Put $a_t = t - 1$, $c_t = t + 2$ and let $F_\sigma(u)$ count genuine new parameters of parity σ in $(a_t, u]$. Partition this interval into at most three intervals of length at most one. The local law, with its endpoint estimate, gives uniformly for $a_t \le u \le c_t$

$$F_\sigma(u) = \frac{\nu(N)}{24}(u^2 - a_t^2) + E_\sigma(u), \qquad \sup |E_\sigma| \ll_N t(\log t)^{-1/20}.$$

Indeed replacing the linear main term on each subinterval by the integral of $\nu(N)r/12$ costs only $O_N(1)$. Integration by parts with $H = h^2$ bounds the remainder by

$$\left|\int_{(a_t,c_t]} H\, dE_\sigma\right| \le \sup|E_\sigma|\big(2\|H\|_\infty + \mathrm{Var}(H)\big) \ll_N t(\log t)^{-1/20}.$$

An eigenvalue at an endpoint is absorbed by the same estimate. Moreover $(\log t)^{-1/20}/\varepsilon \le L(\log t)^{-1/20} \to 0$; the total-spectrum remainder is smaller still. Thus these errors are $o(t\varepsilon)$.

On the complementary real spectral line use $C_M(s,1) \ll_M s+1$, which follows by differencing Risager's refined global formula with remainder $O_M(s/\log s)$. It also bounds each parity and new/sign subspace. The R-scaled Schwartz tails then give $O_A(tR^{-A}) + O_A(t^{-A})$ outside $[t-1, t+2]$, which is $o(t\varepsilon)$, for example with $A = 3$, since $R\varepsilon \to \infty$. The weaker elementary bound $O(s\log s)$ alone would not suffice for this step at the present very slowly growing R. The finitely many imaginary parameters contribute $O_A(e^{CR}t^{-A})$ and hence do not affect the formula.

Order of the arithmetic and analytic parameter choices. The regular residual filter can be chosen with any fixed $0 < \beta < 1$, with excess trace at most $C_\beta(\log X)^{-\beta}$ and total finite-branch complexity at most $\exp(\exp(C_\beta(\log X)^2))$. Here β is the exponent denoted by b in the unit-width excess proposition; it is independent of the window buffer b chosen at the beginning of this subsection. Choose

$$X = \exp(c\sqrt{\log\log t})$$

The constants in the residual filter and trace comparison are fixed after N and β are fixed. Choose $c > 0$ so small that $C_\beta c^2 < 1/2$ for the constant in the double exponential in (4.14), enlarging that constant first to cover all the finitely many sign sectors. Thus

$$\eta = (\log X)^{-\beta} = c^{-\beta}(\log\log t)^{-\beta/2}, \qquad e^{CR}\eta = \exp(-\tfrac{\beta}{2}L + o(L)).$$

The last expression is $o(\varepsilon)$, not just $o(1)$: using $\varepsilon^{-1} \le L$, its ratio to ε is at most $L\exp(-\beta L/2 + o(L))$, which tends to zero. The remaining trace error obeys

$$e^{CR}\sqrt{t}\exp\big(\exp(C_\beta(\log X)^2)\big) \le \sqrt{t}\exp\big(CR + (\log t)^{1/2}\big) = t^{1/2+o(1)} \le t^{3/4}$$

for sufficiently large t. Its ratio to $t\varepsilon$ is at most $Lt^{-1/4} \to 0$. These inequalities explain precisely why the wave norm loss can be absorbed at the stated triple-logarithmic scale. Consequently the counting lemma, the weighted wave majorant, and the proved excess trace estimate give

$$\#\mathscr{E}_N([t,t+\varepsilon]) \le \sum_{r_j \in \mathrm{new}(N)} h(r_j)^2 + Cte^{CR}(\log X)^{-\beta} + O(t^{1-c_0}) \tag{6.2}$$

for a fixed $c_0 > 0$, after summing the finitely many ramified and Fricke sign sectors. For example one may take $c_0 = 1/4$ after decreasing c. This proves the upper asymptotic. The quantitative LV lower bound supplies the reverse inequality. The argument works at fixed parity with its corresponding leading constant.

Proof of Theorem 1.2. The case $N = 1$ follows from the direct modularity argument above and the genuine spectral-window estimates. Suppose $N > 1$. Fix the parity, and work separately in each of the finitely many Fricke and ramified sign sectors. Rebuild the original residual filter using the compressed nonzero-channel reconstruction and the parity equation. It fixes every actual solution exactly. The target metric identifies its exact target with the corresponding cuspidal new/sign space, with constants independent of t.

Apply the counting lemma (4.10) to any finite linearly independent set of solution data in the window. The weighted wave majorant and (4.14), with the parameters in Section 6.1, bound its size by the exact target trace of h^2 plus $o_N(t\varepsilon)$. Distinct Euler products at the same parameter are linearly independent by the argument in Section 2; eigenspaces at distinct nonnegative parameters

are linearly independent because their Laplace eigenvalues are distinct. The uniform bound for all finite families proves finiteness and bounds the full solution count.

Sum the resulting inequalities over the sign sectors. The exact target traces sum to the genuine new trace at the fixed parity; (6.1) gives $\nu(N)t\varepsilon/12 + o_N(t\varepsilon)$. The quantitative LV new-spectrum lower bound and the inclusion of normalized genuine newforms among Euler products give the opposite inequality. This proves the fixed-parity assertion and hence its sum over parity. Finally take, for example, $\varepsilon(t) = (\log\log\log t)^{-1/2}$. The count at $r = t$ is bounded by the count in that window and is therefore $o_N(t)$. □

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